

BCG

ASHUTOSH SHARMA

Round 1 (Technical – 15 mins – Online)

Questions on Regression and Time series project

What is stationarity in the Time series and why is it necessary?

What is multi-collinearity? How will you handle it?

What is omitted variable biasedness?

What is BLUE property?

Few Simple and Basic probability related problems – Total probability theorem, Conditional probability questions

Hypothesis testing, Distribution related questions

Round 2 - (30-35 mins)

Explain one project – (Mainly the Panelist was interested in the project knowledge we have)

Questions on Data engineering work experience – Mainly on cloud technologies

In depth questions related to projects that were done in the job

At end of interviews – Panelists will ask you if you have any questions. Be prepared for it. My panelist was one of founding member of BCG Gamma. So, I asked what changes you saw in BCG Gamma since its founded and something related to it and He was impressed with this question

Round 3 (Case oriented)

Simple stat based questions were asked like p-value, confidence interval etc.

Asked about 1 project

Case was given – To sell a diet plan for weight loss. How do you market it? We had a discussion on that like I gave some ideas, He asked me to calculate the market size and giving hints on How can I approach the problem.

Round 4 (Case based round)

I don't remember the case

Raghavendra

Round 1: Technical

- Asked to mention one project you are proud of.
- Detailed discussion on all projects.
- Focused on NLP topics, including Word2Vec and its inspiration.
- Questions around the candidate's favorite topic (Classification):
 - Concepts and practical applications of Classification.

Round 2: Quantitative Problems

- Scenario-based question:

Let's say the ratio of money the bank has in savings & fixed accounts is $x:y$

$\Rightarrow$ After an year, an average user of the bank will (and has planned to have...)
 $1.05x : 1.07y$ ratio of liquid & frozen money.

Since the average customer is happy with this split, any future investment should also maintain the same ratio.

Let the ratio of new investments be $x':y'$

$\Rightarrow \frac{1.07x'}{1.09y'} = \frac{1.05x}{1.07y}$

$\left((a \times 100)\% \right)$ of money shifts from Savings to fixed.

$\frac{x'}{y'} = \left(\frac{1.09 \times 1.05}{1.07^2} \right) \frac{x}{y} = 0.99965 \frac{x}{y}$

$\frac{x(1-a)}{y+ax} = 0.99965 \frac{x}{y}$

$1-a = 0.99965 + a \frac{x}{y}$

$a = \frac{0.00035}{(1+x/y)}$

- - Will people shift money from one account to another?

Round 3: Business Case

- Warehousing data-related problems.

Aniket Sinha's

Round 1 (30–35 mins)

1. **Resume Discussion:**
 - Detailed questions on every keyword in the resume.
 - Work experience: Power BI and SQL.
2. **Technical Questions:**
 - ROC-AUC curve: In-depth explanation with classification-specific interpretations.
 - TPR and FPR: Interpretation of charts and metrics.
 - Discussed features and target variables from projects.
3. **Project Workflow:**
 - Step-by-step approach to projects, looking for specific pandas functions.
 - Functions for one-hot encoding and train-test split.
 - Next steps after splitting the data.
4. **Case Study:**
 - *Raymond Stores:*
 - Shut down 10 stores using data and business analysis.
 - Segment stores, collect customer data, evaluate nearby stores (cannibalization), and assess profit data.
 - Analyze if shutting the bottom 10 stores by profit is the best approach (consider demographics, assign scores to features).
 - *Behavioral:* Asked questions about handling situations when clients are unsure about their needs.

Round 2 (Partner Round, 40 mins)

1. **Behavioral Questions:**
 - PwC-related responsibilities and project experiences.
 - "If speaking to a company partner, how would you introduce yourself?"
2. **Case Studies:**
 - **Case 1 (US Hospital):**
 - Problem: Patients weren't picking up prescriptions from the pharmacy.
 - Solution: Identify reasons, package solutions based on models.
 - Type 1 & Type 2 errors: Criticality and follow-up questions.
 - **Case 2 (Grocery Setup):**
 - Problem: Loss in revenue.
 - Solution: Time series forecasting for revenue and inventory management.
 - **Case 3 (Telecom Industry):**
 - Problem: Increasing churn rate and reducing profit.
 - Solution: Focus on profit components (costs and revenue). Identify features impacting churn.
3. **Behavioral:**
 - Asked, "How much do you leverage data science for problem statements in reality?"

Round 3 (MD & Partner Round, 15–20 mins)

1. Behavioral and Case-Based:

- Getting-to-know-each-other session (5–7 mins).
- Case: Selling a school. Determine the price by calculating revenue, extra revenues, and costs (non-traditional methods).

2. Interview Style:

- Peaceful, focused on out-of-box thinking and communication.
- Hints provided when needed.

Summary:

- **R1:** Peaceful.
- **R2:** Stress round.
- **R3:** Peaceful, tested creativity and communication.

Awadhesh Mahato :

Online Test

There were 11 questions, duration 90 minutes
8 MCQs (Probability and ML concepts)

One Classification/Regression (two sets were there) problem having 3 parts. Jupyter Platform was given to code.

- 1) Data Manipulation (using Pandas), joining tables, finding aggregate values, changing data types, making one dataframe from these multiple sources
- 2) Data preprocessing, Data splitting, imputation
- 3) Model Fitting and model evaluation

Round 1: Technical

Rate yourself in proficiency in Python, Why so?
Data Structures types and when to use which one
Terms mentioned in CV like, AUC-ROC, Ljung Box statistic
Assumptions of linear regression

Round 2: Case: Dynamic Pricing in Grocery (Let's say Blinkit)

Asked clarifying questions
What's the target variable?
Isn't there upper limit to price, due to price elasticity?
Terms mentioned in CV

Round 3: FMCG Company declining/flat revenue

Asked clarifying questions about company's products; timeline and quantum of the issue
Explained Revenue framework
Got to know that accessibility was an issue

FMCG was giving uniform discount to retailers hence no particular incentive for them to showcase it's products on prime shelves
Need to find right set of retailers and give huge discount and come up with agreement for positioning on prime shelves

VARUN BHAGWANI

Test:

MCQs and 3 data manipulation questions in Pandas.

Round 1:

Introduction

Previous Job related questions.

Some data structures I have worked on.

Regression related questions from my project.

Discussion on my Credit Default Project, a classification project I did in FRM (IIM Calcutta).

Abhishek Anand

Round 1

1. Introduce yourself.
2. How has your experience in PGDBA been?
3. What are your favourite courses in the three semesters?
4. Discussion on work experience – how did you ensure users are properly allocated to the two groups in A/B testing? How can you check if the two groups are truly homogeneous? Can you think of any statistical test to do that?
5. Case study: Flipkart wants to run an A/B test to figure out whether higher prices work for washing machines.
6. Do you have any questions for me?

Round 2 (Partner Round)

1. Introduce yourself.
2. Tell me three interesting facts about Ramayana. Which city is named after Lord Rama's son?
3. Walgreens pharmacy case study: When a patient walks into a doctor's clinic, the doctor writes a prescription. One copy of the prescription is sent to some pharmacies. The patient gets a list of pharmacies and chooses the most convenient one. After the patient chooses, the doctor sends a message to the pharmacy. The pharmacy packs the medicines, but patients don't show up.
4. Follow-up questions: How will you build a model to help solve this problem? How will you explain ROC-AUC to the CEO?
5. Do you have any questions for me?

Round 3 (MDP Round)

1. Where are you from?
2. The Government of India wants to sell Durgapur station. It has asked BCG to figure out what the ideal selling price should be for an investor.
3. How many passengers travel via Durgapur railway station in a day?

Round 4 (Partner Round)

1. Case study 1: An Insurance Company's CEO asks BCG to help increase his company's sales/revenue.
2. Case study 2: BCG is working for a fashion retailer. Sales are declining or stagnant. Help them.

Madhur Rawat

Exam

- MCQs + 3 coding questions on:
 - Data manipulation using Pandas.
 - Building a basic ML model.
- Internet was allowed.

Pre-Round 1: Project-Based Discussion

- Basic questions on projects and models used.

QRT

ASHUTOSH SHARMA

Technical Test - 2 coding questions (Some questions that can be solved using Hashing), 10 stats and ML based MCQ questions – 45 mins (Difficulty level – Moderate)

Round 1 (15 -20 mins) – On campus (1 Panelist)

Few questions on Regression project

How do you cut a cake in 8 equal parts? (Standard GFG puzzle)

Checked my knowledge on Finance

Explain about CAPM model?

Case – How do you calculate the coal demand for the country?

SAHIL CHADHA

Test round

Test is just after ppt - on laptop - in offline mode (invigilated and proctored). There were 2 coding questions and other MCQs that were related to Statistics - probability, hypothesis testing - Chi square test, KS test, T-test, ML - regularisation, , . Coding Questions - DSA - 1st question was based on a 2 pointer approach. 2nd Question - no idea (refer Atharva / Madhur, Nikunj). (12 people cleared this round)

Interview Round 1

12 students got shortlisted. The 1st round was based on intro, CV - projects, Regression assumptions, puzzle - how to cut cake into 8 equal pieces in 3 cuts. Question related to properties of normal distribution, skewness - mean median mode in right skewed distribution. Then they asked me to think of some unconventional way to get signals to invest in the market. Studentised vs standardised residuals. Why QRT. duration - 30-40 min.

Interview Round 2

Interviewer was from Europe (online) (Emmanuel) . 1st he asked for my intro and background. It was a strategy-based case. Given price data of oil(crude) and gasoline, how can we make a trading strategy? What factors to look into. He was leading the case and was giving hints. Discussions were on topics like pair trading, heading the position, etc. At the end, he asked a question, given 2 portfolios with 0 sharpe ratio and other with 1 sharpe ratio, how to combine both to improve sharpe ratio. Then he linked this with the given case. This round went for around 40 min, and

Interview Round 3.

For this round, only 4 people got shortlisted. Different interviewer from Europe (online) (Robin). He first introduced himself. He was head of commodity investment. Then he explained about the role a little bit. Then he asked me about my background and why I am interested in this role. Then the discussion went on one of my projects which was on portfolio optimization. I explained our approach, but he was not satisfied, and he asked why we didn't use the optimization route and closed form solution. From thereon, it went a little mathematical, and he was asking questions on the density of a set of invertible matrices,

which I didn't have any idea of, but he wanted to have a discussion on that. This discussion didn't go well for me.

Soham's Interview Experience

Round 1 (~30 min)

1. You have 12 balls, 11 identical, and one odd (heavier or lighter). You can use a weighing balance three times. Determine the odd ball and whether it's heavier or lighter.
2. For your petrol price time series project, if you had access to any data, what exogenous variables would you use, and why?
3. What is the area covered by three standard deviations of the normal distribution?
4. How would you handle skewness in your data?
5. Name a statistical test to detect a leverage point in regression. Are leverage points and outliers the same? How would you treat them?
6. What is the difference between SVM and the maximum margin classifier?
7. In your classification project, you handled multicollinearity by regressing one highly correlated feature on another and using the residual. Might this cause bias? What alternatives could you use?
8. If you have two models—one complex with good validation metrics and another simple but with worse metrics—which would you choose and why?
9. What is the PMF of the Poisson distribution?
10. In a decision tree, at what level does the splitting occur? What criteria would you use to split nodes?

Peehoo Jain

Round 1 (Offline, 30 mins)

1. **Regression (ISLR):**
 - Lasso and Ridge.
 - Multicollinearity.
 - Autocorrelation (ACF Lag 1, 2, ...).
2. **Case Study:**
 - Stock market trading: Strategy to determine rise/fall.

Round 2 (Online, Germany-Based Panelist)

1. **Case Study:**
 - Oil and gas prices: Quantitative research and case study discussion.
 - *Note:* No puzzles were asked in this round, though they are usually part of the interview.

Round 3 (HR)

1. **Behavioral Question:**

- Where do you see yourself in 5 years?

Premjeet

Round 1 (Doesn't Remember Much)

1. **Puzzles:**
 - Math + Probability.
2. **Monte Carlo Simulation:**
 - How to estimate the value of π using the Monte Carlo method.
3. **Casino Strategy:**
 - Scenario: Playing roulette in a casino. What would your strategy be?

Parth Pandey

QRT:

Test Round:

2 coding questions on basic DSA regarding time complexity.
10 MCQs on Statistics / ML.

1st Round:

Probability: Minimum number of shirts and pants required for 365 unique combinations.

Tell me about LSTM, its mathematical equation

Tell me about your Regression project. What is the difference between Lasso and Ridge regression, and other details about the project?
(My project was regarding the prediction of quantity of oil imports.)

Poisson Distribution - its PMF and discussion about its parameter with a real life example.

Atharva Kulkarni

Online Test

- Basics of machine learning, regression, regularization, overfitting, and underfitting?
- Quant-based problems on profit/loss and speed-distance?
- z-table to figure out probability in statistics?
- Coding question
- Swap positions of individual elements of an array based on a rule until the original sequence is restored? How many steps are required?

Round 1

- Are you quick at mental math? (Asked to multiply 33×29 and 43×41)
- Can you introduce yourself, focusing on why you chose PGDBA and what you liked about it?
- Why do you want to join QRT?
- How do you cut a cake into 8 equal parts with just three cuts?
- How does the mean, median, and mode behave in a right-skewed distribution?
- Can you explain your regression project?
- Do you think all properties in San Francisco are priced higher than those in New York?
- How would you use latitude-longitude information to estimate property prices?
- What do you understand by multicollinearity? How does it affect a model?
- What are the assumptions of Linear Regression
- What were the top three important features in your regression project?
- What do you understand about SHAP values? How do they differ from interpreting regression coefficients?
- If you were to build a buy/sell indicator for a stock or index, what unconventional variables would you incorporate?
- What do you understand by the stationarity of a time series?
- What are the three sub-types of the ADF test?
- Where do you see yourself in the future?
- Do you have any questions for me?

Round 2

- Suppose you have oil as a commodity. How would you formulate a strategy to earn from it?
- How do you decide if you should take a position in oil futures?
- If you had access to previous oil prices for 10 years, how would you use this data to model future prices?
- How does demand and supply affect oil prices?
- How would you model the status of oil pipelines and trade routes to impact pricing?
- How does a non-functional pipeline or trade route impact oil prices?
- How would you use a graph network model for oil depots and pipelines?
- If you did not have access to any exogenous variables, how would you build a buy/sell/hold indicator for oil price forecasting?

- Would SARIMA or XGBoost be better for time series modeling in this context? Why?

Round 3

- Can you introduce yourself and walk me through your resume?
- What do you understand by discretionary equity assets of Fidelity Investments?
- What do you understand by 'Systematic Investment Strategy'? How does it differ from discretionary strategy?
- What was your team's role at Fidelity, and how did your work impact the firm?
- What did your team do at Morgan Stanley?
- What is the Prime Brokerage (PB) business of Morgan Stanley? How does it generate revenue?
- How did your team support Morgan Stanley's PB services?
- Who were the end-users of the software applications you worked on?
- If given the opportunity, how would you apply AI/ML in your previous workplace? What features would you collect, and how would they be useful?
- How comfortable are you with regression analysis?
- How would you explain regression to a layman?
- How proficient are you in coding?
- How would your AWS certification and experience with cloud migration help in this role?

Round 4

- Can you tell me about yourself?
- Can you share details about your work experience?
- What motivates you to join QRT?

Madhur Rawat

Exam

- Coding Questions + MCQs.

Round 1:

- Calculation of π .
- Real-life application of Exponential Distribution.

Round 2:

Discussion on Crude Oil Trading:

- Factors affecting crude oil prices.
- Key indicators for price movement.
- Asked to create a trading strategy.
- Why do stock prices change over time?
- HR Questions: Same as previous rounds.

Namit Pulyani

Exam

- Coding questions + MCQs.

Round 1:

- Puzzles related to the Roulette game: Given 50 Red, 50 Black, and 1 Green slot, choose a strategy to maximize wins. Follow-up questions on this puzzle.
- How will you calculate the value of π ?
- Give a real-life example of Poisson Distribution.

Round 2:

- How will you estimate the price of corn?
- Why do you want to join QRT?
- What are your long-term career goals?

Nikunj Gupta

Test Round

- Given a permutation that transforms an array $[a, b, c, d]$ to $[a, d, c, b]$, determine how many times it must be applied to return to the original array.
- Difficulty Level: Easy - Moderate.

Round 1:

1. Assumptions of Linear Regression?
2. What happens to coefficient variance in LR if heteroscedasticity is not fixed?
3. Comparison of coefficients:
 - $b_1 > b_2$, $b_1 = b_2$, or $b_1 < b_2$?
4. Difference between Gini Index and Cross-Entropy?
5. What is Poisson Distribution?
6. Basic probability problem.
7. Prevalence = 0.01, Sensitivity = 0.95, Specificity = 0.98 → Find FP/TP ratio.
8. Puzzle: 2 ropes, burn at non-uniform rates. Each burns completely in 1 hour. Measure exactly 45 minutes.
9. What is RNN? Advantage of Transformer over RNN?
10. Favourite ML algorithm?
11. finance-related ML project from CV. Explain your project.
 - Why these features?
 - What is the Sharpe Ratio?
12. What is SARIMA? Disadvantages?

Round 2:

1. Why did you choose to work for AB InBev after graduation?
2. How do you forecast the price of corn in the market?
3. Trained an ML model on 10-15 years of data. After deployment, performance starts dropping. How do you fix it?
4. Long-term aspirations?

P&G

SAHIL CHADHA

PnG test - it's a psychometric test followed by speed maths and reasoning. Do watch youtube videos before giving the test. There's a strategy to be followed in the test.

Subhajit Tarafdar

Round 1 (15 mins)

Basic intro

2 Projects –

- Regression Basic questions
- Mtech thesis (Bio Medical) – CNN – Basic Questions
 - o What did you do?
 - o How did you do?
 - o Have you done any deployment before?

Some practical questions

How my modelling will help?

What result did you get?

Ishan

Round 1

1. Basic ML Questions:

- o Decision Trees:
 - Complete explanation.
 - Greedy approach.
 - Pruning: Benefits and drawbacks.
- o Ensemble methods.

2. Technical Questions:

- o Shared screen: Explained a code snippet and filled in missing parts.
- o OOPs questions: Python classes, overwriting, etc.
- o Git commands.

3. Wrap-Up:

- o "Do you have any questions?"

Round 2 (Leadership Round)

1. Leadership question: Discussed experiences showcasing leadership skills.
2. Interview Style: Frequent interruptions and probing during answers.

Manas

Round 1 (Technical, 30–35 mins, Panelist from Singapore)

1. **Introduction:** Overview of coursework and projects.
2. **Used Car Price Prediction:**
 - Why was this project chosen?
 - Impact and monetization of the model.
 - Ensuring the robustness of the model.
3. **Linear Regression:**
 - Assumptions (e.g., multicollinearity, homoscedasticity).
 - Use of Ridge and Lasso regression.
 - Explanation of feature selection and justification.
 - How would you sell this model to a client?
4. **Portfolio Optimization:**
 - Hidden Markov Models: Why were they chosen?
 - Advantages and limitations of this model.
 - Relevance of Modern Portfolio Theory (finance concepts).
5. **General Questions:**
 - Life of a Data Scientist at P&G?

Round 2 (Tech + Case Study + Guesstimate, 40 mins)

1. **Introduction:** Projects overview.
2. **Case Study:**
 - Scenario: Ads for brands on social media platforms (YouTube, Facebook, Instagram).
 - Problem: Sales have decreased; model accuracy is poor.
 - Constraints: No consumer-level data, only e-commerce data.
 - Business considerations: Festivals (e.g., Big Billion Day sales).
3. **Guesstimate:**
 - How many planes travel between Kolkata and Mumbai on Diwali?
4. **Interactive Approach:**
 - Asked clarifying questions to define the business needs.

Round 3 (AMA Region Head)

1. **Introduction:**
 - Discussed a project close to the heart: Drug-Drug Interaction (data sourced from Drug Bank).
 - Problem: Patients taking drugs for chronic diseases → next drug for different diseases.
2. **Technical Questions:**
 - Features used in the project.
 - Model used: ANN. Why not XGBoost?
 - Ensuring model performance and handling data drift.
 - Monetization strategies and managing structural data changes.
3. **HR Questions:**

- Conflict resolution.
 - Managing strict deadlines.
 - Accountability when a model fails.
 - Motivating someone to act against their will (structured CAR framework: Context, Actions, Results).
4. **Behavioral Questions:**
- Explaining technical concepts in simple terms.
 - Leadership skills and documentation ability.
 - Judged on communication and framing thoughts rather than deep technical skills.

Muskan Agarwal

P & G :

Pre-Round:

- Online Psychometric Test (MCQ Format)
 - Includes ethical questions, digit challenges, ratings, and pattern challenges.
 - Tip: Watch YouTube videos on P&G Assessment Test for better preparation.
- CV + Test-Based Shortlisting

1st Round: Technical Round

- Introduction
- Discussion on my project using XGBoost
 - Detailed questions on XGBoost and its advantages over other ML algorithms.
- Few questions on:
 - Object-Oriented Programming (OOP) in Python
 - SQL
- Tip:

- Present answers in a structured way.
- Review all past projects thoroughly.

2nd Round: HR / Ethical Round

- Power BI Discussion (since I used it in my previous job)
 - Asked about advantages and disadvantages of Power BI.
- Behavioral Questions using the C-A-R format:
 - C (Context), A (Action), R (Result) framework for structured responses.
 - Example:
 - What revolutionary step did you take in your last job, and what did you learn?
 - What benefits did you gain from PGDBA?
 - How will you contribute to P&G? Why should we hire you?

3rd Round: Case Study & Discussion

- Difference between a Data Scientist and a Data Engineer
 - In-depth discussion on roles and responsibilities.
 - Asked how I would resolve conflicts between them as a Chief IT Officer.
- Case Study:
 - Scenario:
 - As Chief IT Officer of P&G, my team developed a sales forecasting model for a new shampoo launch in Dharavi, Mumbai.
 - A competitor also launched a similar product simultaneously.
 - Due to this competition, the sales team is failing to meet the forecasted sales targets.

- The sales team requests the removal of the model as they believe it is inaccurate.
- Question: What will you do?
- Tip:
 - Ask detailed clarifying questions before answering.
 - Consider as many factors as possible in your response.

Overall Tips for P&G Interviews:

- Research P&G thoroughly:
 - Understand the company's mission, vision, product categories, and net worth.
- Practice structured responses using the C-A-R framework.
- Be prepared for both technical and behavioral questions.
- Think critically in case study discussions.

PARTH PANDEY

Pre-Round:

Online Psychometric Test (MCQ Format)

Includes ethical questions, digit challenges, ratings, and pattern challenges.

Tip: Watch YouTube videos on P&G Assessment Test for better preparation.

Round 1:

Introduction.

Detailed discussion on my classification project (Logistic Regression). He wanted me to explain most of the things mathematically. (I derived the loss function there.)

He asked me about any non parametric algorithm I know for classification.
(My answer was Decision Tree.)

Then a detailed discussion on Random Forest and how it's different from Boosting. Lastly we discussed XGBoost.

Round 2:

What innovative work have you done in your last job? Some follow up questions.

What motivates you to join P&G?

How do you stand out from other candidates?

Round 3:

Some questions were repeated from the previous round?

Detailed discussion on my Time Series project on Foreign Tourist arrival, its business application.

VARUN BHAGWANI

Pre-Round:

Online Psychometric Test (MCQ Format)

Includes ethical questions, digit challenges, ratings, and pattern challenges.

Tip: Watch YouTube videos on P&G Assessment Test for better preparation.

Round 1:

Introduction.

Favourite Project - Regression. Discussion on RMSE and Adjusted RMSE.

Lasso/Ridge - Which of them can be used for feature selection? Why?

Discussion on my DL project on PII (Personal Identifiable Information) - any word or text that can give some information about a person, for example, name, address, phone number etc.

Round 2:

Behavioural Questions:-

Instance where you have influenced/ led a team. Cross questioning on what I did and why?

Any big hurdle I came across that has changed me.

If they are not satisfied with the answers, they will ask another question.

Himanshu Singh

Shortlisting

- Psychometric test: A set of over a 100 personality-based questions.
- Aptitude tests: Memory and fast calculation games, similar to cheap online mind games.

First Round

- Introduction.
- Asked to discuss one project of choice.
 - Discussed a project on Sentiment-Based Product Development.
 - Provided a detailed walkthrough of the project, with logical and technical questions throughout.
 - Follow-up questions on other projects and highlighted points in the resume.
 - The focus was more on intuitive understanding rather than mathematical intricacies.
 - If unsure about a topic, the interviewer was open to providing hints.

Second Round: Case

- Given a case study related to identifying factors affecting product sales.
- Questions on methodology: which techniques to use, what features to employ, and how to source data.
- Discussion on model effectiveness under anomalous conditions (e.g., COVID-19).

Third Round

- Increasingly complex situational questions.

- Started with a scenario involving deadlines (e.g., deadlines in a week, month, six months, and a year).
- Added constraints like being assigned a critical task by the VP within a fortnight.
- Successive constraints: No help from colleagues, no access to certain resources, inability to negotiate, immediate execution required, etc.

Fourth Round: Personal Round

- Introduction, followed by personal questions about family and life.
- Asked to discuss a project of choice with counter-questions.
- Asked to describe an incident where I had gone "against the grain."

Madhur Rawat

Pre-Round 1: Technical Round

- Brief overview of:
 - Statistics
 - ML methods
 - DL basics

Pre-Round 2: Behavioral Round

- Given various situations, asked about next steps and decision-making.

Nilkunj Gupta

Psychometric Test:

- Leadership skills & passion.
- Scale: 1-4 rating system.

Technical Round:

1. Basic ML, DL questions.

Stress Interview:

- Designed to pressure-test candidates.
- Case Example
 - Scenario: CFO asks which sales-related projects you can deliver in the next 2 weeks.
 - how will you adjust to different scenarios (asked by interviewer)
- They keep pushing back on your answers

Round with IT Global Head:

General discussion, storytelling, passion, creativity

Time Series Project Discussion:

- Sales of household electronic items in the UK & COVID impact.
- SARIMAX model: Finding optimal order parameters.

Rituja Metar

Round 1

1. Explain the classification project.
2. How does the Node2Vec algorithm work?
3. Explain precision and recall. In this case, which metric was more important and why?

4. Provide an example where precision is more important and another where recall is more important.
5. When is accuracy not a good measure to evaluate the performance of a classification algorithm?
6. Explain your NLP project.
7. Describe the architecture of the BERT model.
8. How did you implement the BERT model?
9. What are RNNs and LSTMs?
10. What shortcomings of RNNs does LSTM solve?
11. Why is transfer learning used?
12. How is transfer learning applied in CNN and NLP related tasks?

Soumendu Mandal

Screening

- Aptitude and reasoning-based elimination round.
- CV-based shortlisting.

Round 1:

- Introduction.
- Asked about favourite project
- Discussion on project details and CV points.
- Precision vs. Recall—given cases where one is more important than the other.
- Questions on XGBoost, data outlier handling, and imputation methods.

Round 2

- Given a scenario: Four pending projects with different timelines + urgent CFO request.
- How would you prioritize and handle the workload?

Providence

SAHIL CHADHA

Test: its MCQ based test. There are sections - Deep learning, probability, ML, Stats and inference. Bayesian nets were also asked.

Pre - Round 1 (online)

In this round he expected me to tell the name of specific terms. He asked about likelihood, clustering / data segmentation etc. He gave one question - what would be the activation function on the last layer of a neural network if output is continuous variable. (ans - no activation function required)

He asked about data drift and how would you test it if data is categorical (Chi square test) and continuous (can do KS test etc).

After these technical questions, he asked me to write a python code. Given a list: ["chennai", "delhi", "chennai", "bengaluru", "pune"] - make a dictionary with { city_i : (a,b,c)}

Where city_i is ith element of list, a is vowel count in that city, b is consonant count in that city, and c is the frequency of city occurred in list.

RAHUL SINGH

Screening – CV based Shortlisting

Then Test (Linear Algebra, Deep Learning, Probability)

Then **1st round interview** (29 Students were there)

Intro

2 coding ques-

Two lists are there, find the common element & pass it into the list

Write code to create binary tree, something like this.

Probability qsn – There are 40 different shipments out of which 12 are defective, We are sending eight shipment out of country. Then what is the probability that at least three shipments are defective?

Case Study – Out of 10 million population, you need to find that 1% population who is going to take the admission into your hospital. So find the target audience.

2nd round Interview-

Intro

Work Ex

Tell me the flow of NLP (1980-2022 journey).. From rule based to transformer

Coding ques –

Find the unique element between 2 strings. Its time complexity, how u would optimise this.

SQL query

Case Study- You need to design a new website, which looks like linkedin. So what are the various factors you will consider? And how data analytic can be applied into it

Then again workex

TIPS- Interview level varies from panel to panel. Read whatsoever u can read. Coding ques will be asked definitely, SQL also, Also the DSA will be asked, Be ready for Bayesian Statistics, Probability

MANIKYAM

Round 1

1. *Regression and MLE:*

- Derive the equation for Maximum Likelihood Estimation (MLE) in regression.
- For large datasets, which method is better for estimating coefficients—OLS or gradient descent? Why?
- Explain the computational complexity of OLS when handling large datasets.

2. *Ridge and Lasso:*

- What are Ridge and Lasso regression, and when would you use each?
- Explain their differences mathematically and conceptually.

3. *Cross-Entropy Loss:*

- What is the cross-entropy loss function?
- Write the code to implement it.

4. *Transformers:*

- What is the shape of the attention matrix in transformers?
- How does attention work in transformer models?

5. *Probability Distributions:*

- Can you explain different types of probability distributions and their applications?

6. *Gradient Descent with Momentum:*

- What is gradient descent with momentum, and why is it used?
- How does it differ from standard gradient descent?

Round 2

1. *Deep Learning Project Walkthrough:*

- Walk us through one of your deep learning projects.
- Beyond confusion matrix metrics, how would you confirm that the model is learning the correct features in an image? (Hint: Techniques like occlusion analysis.)

2. *Batch Normalization:*

- What is batch normalization, and how does it improve deep learning models?

3. *Case Study:*

- How would you predict the percentage of the amount that will be refunded after an insurance claim is filed?
- Additionally, how would you estimate the time required for the claim to be processed?

Raghavendra

Round 1

- No discussion around profile or projects.
- Questions focused on basic Deep Learning (DL) concepts:
 - How comfortable are you with DL?
 - Types of loss functions and their use cases.
 - Dropout rate: syntax and parameter names.
- Perplexity: What is it?
- Transformer Architecture: Attention and Self-Attention.
- Numpy array modification questions.
- Definition of all classification metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score

Surya Kiran

Pre-Round Interview (Online)

- Focused on stats basics and coding skills.
- 1. What's a PDF?
- 2. Practical applications of the exponential distribution.
- 3. State Chebyshev's inequality.
- 4. What is backpropagation?
- 5. State different parameter optimization techniques in neural networks (e.g., stochastic gradient descent, momentum methods).
- 6. Write a code to merge a list of lists into a single list (screen sharing required).

Second Round (On-Campus with Dr. Udai Shankar)

1. Questions on stack and queue insertion/deletion time complexities.
2. Why are principal components in PCA orthogonal?
3. What is stationarity in time series?
4. Case Study: Predict diabetes occurrence from random health records of patients.
5. Is predicting Rohit Sharma's score in the next match a good data science question?

Ishan

Round 1 (Online Interview)

1. **Work Experience:** Questions related to prior work experience.
2. **ML Questions:**
 - Classification: Precision vs. Recall, examples of when to use each.
 - Undersampling vs. Oversampling.
3. **Coding Questions:**
 - Logical coding question involving loops and if-else conditions.
 - Forming triplets.
 - Recursion-based question.
4. **SQL:** Problem-solving with SQL queries.

Aniket Sinha

Coding Questions:

- Without loops, find common elements in lists using set functions.

Puzzles:

- Poisson distribution puzzles (similar to SPA notes and textbooks).

Peehoo Jain's Interview Experience

Pre-Round

1. **Coding:**
 - Python: Recursion-based question (Fibonacci series, hints to solve in one line).
 - SQL: Basic queries, including joins and aggregates. (*Leetcode 50 interview questions recommended*).

Round 1

1. **Resume-Based Questions:**
 - Focused on highlighted points.
2. **SQL:**
 - Queries related to filling details, joins, and aggregate functions.
3. **Deep Learning:**
 - In-depth technical questions.
4. **Behavioral:**
 - Teammate not contributing: What would you do?

Manas

Round 1 (Extremely Technical)

1. **Coding Questions:**
 - Flatten a nested list and find the median (`lis(list(list))`).
 - Write a function to find palindromes from a list of strings.
2. **Linear Algebra:**
 - Intuitive explanations of rank of a matrix, positive definite, semi-positive definite matrices.
 - Application of linear algebra in linear regression.
3. **NLP:**
 - Questions based on keywords in the resume.
4. **Clustering:**
 - Techniques and applications.

Round 2 (Easier, Luck-Based)

1. **Project Discussion:**
 - Drug-Drug Interaction Project.

Advice for Providence:

- Strong coding fundamentals (loops, recursion).
- In-depth understanding of linear algebra (intuitive explanations).

- Ability to explain technical concepts in layman terms.

Premjeet

Pre-Round

1. **Technical Questions:**
 - Approximation to normal distribution (specific and precise answers were required).
 - Activation functions: Which to use in the final layer for classification and regression tasks.
 - PCA: Main concept and explanation of eigenvalue decomposition.
2. **Coding Tasks:**
 - Extract tags and their frequencies from a dataset.
 - Create a binary search tree.
3. **Model Evaluation:**
 - How to test if a model is still working (hypothesis testing).
 - Checking for data drift.

Round 1

1. **Case Study:**
 - Business data science case.
 - Created and discussed a fraud detection scenario.
2. **Work Experience Discussion:**
 - Reflections on previous work experience:
 - What changes would you make based on your new knowledge?
 - Technologies worked on (Hugging Face, PyTorch, TensorFlow).
 - Challenges faced with these technologies.
3. **Bayesian Concepts:**
 - Formula: $\text{Posterior} = \text{Prior} \times A/B$.
 - Asked to explain A and B.

Round 2 (Behavioral)

1. **Situational Questions:**
 - Problems faced during work.
 - Conflict resolution.
 - General introduction.

Harsh Raj

Providence (30 Minutes)

1. Interviewer asked me to explain Healthcare project, and then follow-up questions were from that project.
2. Difference between BERT and RoBERTa
3. He gave me an input sequence vector and asked me the dimension of K, Q, V vectors.
4. How is embedding generated in BERT
5. What is Rank of a matrix
6. What is span of a matrix
7. How do you check if the vectors are linearly independent
8. Asked me to write a code for finding the trace of a matrix.

SAMEER GUPTA

1. Pre-Selection Online Assessment:

Questions focused on aptitude, machine learning, coding, and deep learning. The questions tested basic concepts and practical knowledge in these areas.

2. Pre-Selection Interview Round (20 minutes):

Machine Learning: Basic questions about machine learning, such as "What do you understand by machine learning?" and "How do you approach a machine learning problem?" Your response described machine learning as both a science and an art, providing a balanced perspective.

Business Case: You were given a high-level business case involving bank fraud detection and how you would implement a machine learning model for fraud detection.

Probability Distributions: You were asked about the Poisson distribution, and your knowledge of probability distributions was tested.

SQL Queries: Basic SQL queries were asked to assess your database skills.

3. Final Round Interview (Placement Day):

Introduction: You began with your introduction.

Coding Question: A question asked to determine whether a number is prime or not, testing your algorithmic thinking and coding ability.

Business Case Discussion (15 minutes):

The case involved an online SIM card company with extensive customer support interaction data.

You were tasked with designing a solution to build an LLM (Large Language Model) that could respond like customer support, handling interactions effectively.

Key elements of the case:

Data sources and collection methods.

Data cleaning and preparation.

Selection of the appropriate LLM, evaluation of benchmarks, and fine-tuning strategies.

Deployment of the solution and considerations for scaling.

Providence Interview - Key Points:

1. Panel Variability:

The interview experience at Providence can vary significantly based on the panel you are assigned.

Some candidates might have a smooth experience, with the panel asking only basic questions and getting selected easily.

In contrast, other candidates might face a more challenging panel, where they are asked in-depth questions on topics like LLM fine-tuning and deep learning, leading to a much tougher interview.

2. Luck Factor:

The outcome of the Providence interview is highly dependent on luck and which panel you get assigned to.

Even if you're well-prepared and capable, the nature of the questions and the panel's expectations can be unpredictable, which can significantly affect your chances of getting selected.

3. Unpredictable Outcome:

It's important not to overestimate your chances or set too high expectations, as even a strong candidate might not get selected depending on how the interview goes.

Ankush Pandit

Round 1

1. Can you introduce yourself?
2. Would you like to discuss ML or DL?
3. Is supervised or unsupervised learning preferred?
4. What is a loss function, and what are such functions in supervised learning?
5. What is cross-entropy? Can you provide its formula?
6. What is entropy?
7. Give formula and explain how it is used
8. What is information gain? Can you provide its formula?
9. How do you perform Maximum Likelihood Estimation (MLE) step by step?
10. What is Maximum A Posteriori (MAP)
12. If a weight and the final output are far apart in a network, how can it update the final output?
13. What is the vanishing gradient problem, and how does it happen?
14. What are activation functions? Explain ReLU and Sigmoid.
15. How do you calculate the size of the attention matrix based on input dimensions?
17. What are precision and recall? Can you provide their formulas?
18. Is logistic regression a linear model?
19. What is the difference between cross-entropy and softmax?
20. What do you mean by linearity? Can you provide an example of a non-linear function?
21. What is PCA (Principal Component Analysis)?
22. Given a dataset of 100 rows and 10 columns, how would you perform PCA step by step?

Round 2 (Managerial + Behavioral)

1. Introduce yourself?
2. Can you discuss your work experience at Amazon?
3. Can you discuss your work experience at Motorola?
4. What was the importance of your team at Amazon? What used to happen before your team was formed, and what value did you add?
5. Can you elaborate on a specific project you worked on?
6. What were the particular challenges you faced in that project?
7. Now that you understand analytics and data science, how would you incorporate that knowledge to improve the product?
8. You mentioned that three stakeholders benefited from your offering. Can you discuss how each stakeholder was impacted?

9. What kind of data would you collect, and how would you collect it?
10. What kind of model can be used to improve the product?

Round 3 (Culture Fit)

1. Introduce yourself?
2. Can you give an example of a situation where you handled conflicts?
3. Can you discuss certain challenges you faced and how you handled them? (One from work experience, one from PGDBA)
4. Amazon has certain leadership principles. Which ones are your favorite?
5. What is your biggest failure?

Madhur Rawat

Pre-Round: Technical Screening

- Questions on ML and DL methods used in Computer Vision.
- Coding Question: RMSE using NumPy.

Round 1: Project-Based Discussion

- Discussion based on CV.
- Had 3 healthcare-related projects → Discussed:
 - Business impact.
 - Methodology used.
 - New ideas
- Brief overview of other projects and ML/DL methods used.

Round 2: HR Round

- Why Providence?
- Situation-based question: *Your teammate refuses to work—how will you handle it?*

Namit Pulyani

Pre-Round: Technical Screening

- Questions on OLS, Ridge, and Lasso regression.
- Clustering: K-means, DBSCAN.
- PCA-related questions.
- Time series concepts: ACF and PACF plots.
- Pseudocode basic array manipulation
- Basic questions on RNN, LSTM, and CNN (no in-depth discussion).

Round 1: Case-Based Questions

- Case 1:
 - Given full team data, how will you provide recommendations for player and team strategies? (since I mentioned cricket in my introduction.)
- Case 2:
 - Providence plans to launch a cancer drug. Its competitor is performing well, so it needs an optimal strategy. How would you analyze various factors before entering the market?
- Conducting a real-life Z-test.

Round 2: HR Round

- Why do you want to join Providence?
- Have you ever faced conflicts in group projects? How did you handle them?
- Shifted the discussion towards my healthcare-related project.

Rituja Metar

Round 2 Interview

1. Explain Bayes' Theorem and its applications.
2. How would you model a single coin toss
3. What are some key properties of continuous probability distributions?
4. List some activation functions.
5. Explain the sigmoid activation function, formula, and use case.
6. Which activation function is used for multiclass classification, and why? (
7. Explain hypothesis testing. What are one-sided and two-sided tests?
8. What are the key assumptions of hypothesis testing?
9. What is the KS test, and when is it used?
10. Explain the chi-squared test and its applications.
11. What is KL Divergence? Explain its significance and use cases.
12. What is self-attention and cross-attention?
13. A simple coding question on implementing dictionary

Round 3 Interview

1. Given a dataset with variables x_1, x_2, x_3, x_4, y , where x_1 causes x_2 , x_2 causes x_3 , x_3 causes x_4 , how would you model this relationship?
2. What are Principal Components? How are they computed? Are they always perpendicular to each other, and if so, why?
3. Suppose we have a list of events with precedence constraints, e.g., Event A must be completed before Event B, and so on. How would you determine the order in which the events should be completed?
4. How does the topological sort algorithm work?
5. Given patient notes, how would you build a model to extract meaningful insights from the data?

6. Explain the architecture of the BERT model and its key components.
7. What are Gaussian Mixture Models (GMMs)? How would you estimate the parameters of a GMM?
8. How does the Expectation-Maximization (EM) algorithm work?
9. Suppose you are given a task with two possible approaches, and you have a tight deadline

Approach A: Use an already implemented solution that takes less time but may not yield the best results.

Approach B: Use a research-intensive approach that takes more time but has a high chance of delivering the best results.

Which approach would you choose and why?

Games 24x7

AKSHAY K

Test

How would you identify users? – Churn

Aptitude questions – MCQ's

Hypothesis Testing, Probability, SQL – Descriptive questions

An Informal session happened in Tata Hall before placement day.

Round 1 (Case Round) (Panelists – 2 (Both are Batch 1 Alumni))

FADML – Graph Neural Network based project. (Used for classification for Fraud detection)

- Why did you use it instead of other simple models?

Talked about the Use cases of CNN

Compare the performance of GNN vs Random Forest

Cases – My 11 circle, Rummy circle, Fantasy betting

New Advertising campaign – Whom would you target? What mechanism will you follow?

End to End ML project.

(Ans – People who aren't using for longer recently, People who are more likely to come back)

Tip – Ask chat GPT to generate some kind of problems for practice

A/B Testing?

Test statistic?

If the alpha value given, how do you choose sample size?

Round 2

Hypothesis Testing – Answer in simple terms.

Why do you use statistic?

Precision, Recall – when do you use each? What do you do if you want to use both?

Tip for Case study – Ask clarifying questions before starting solution

Round 3

Where do you want to see yourself in 10 years?

Why do you choose Games 24X7?

What kind of work you want to do?

GenAI and Fancy algo coming up, how would you go about dealing with new technologies?

You made recommendation algorithm, how would you differentiate the performance of new environment with the previous one?

Suggestions –

- Do well in tests
- Do cases. look at Games 2X7 website, think about potential issues or models they may have.

SAHIL CHADHA

Test

There are 2 cases and you have to write the solution (type it). Go through the previous pdf of questions, you will get from your PRs. prepare them and listen to their ppt carefully, take screenshots of business metrics shown by them.

Then there is one question based on SQL. again you have to write the solution.

There was one question on the power of the test. (inference/hypothesis testing)

And one probability question. It was an easy one, don't recall the exact statement.

Round 1 -

1st introduction. Then he took my credit card default project. Asked objective and how exactly I did that. Why F beta score, etc. Then he gave me a scenario that I am working in a bank and have access to all this data. How will I approach the same project? How will I collect data and how will I merge them, how will I define the business metric etc.

Then he gave me a scenario, wherein I have to find fraud in the ecommerce business. But this is a fresh project - so you have to define fraud, tell the approach to collect the data, and how to validate it. Caveat is that it has not been done before. Lots of cross questioning happened in this case

Then he asked me some technical questions from my CV (the keywords) and in general. Like when to choose a neural net and when to go for Logistic regression, etc..

Round 2 -

It was sort of a leadership round and compatibility check round. It happened with senior leadership. They explained my vision of Games24, and asked me what I understood about Games and why I wanted to join it. Why I am fascinated about the product. Then they asked me if I would prefer updating myself on cutting edge AI ML stuff or just learn that which is being required by the current project. They were asking questions to understand if I will take initiatives, etc. Then he asked, let's say you deployed a model, but its not working and you are not getting results, i.e. business metrics are not showing success. What will be your approach and how will you handle that?

Then he asked if I had any questions - and then we interacted for some time.

RAHUL SINGH

Screening – Test was taken, case study asked, ML ques asked, some basic coding ques asked

Interview-

How would u rate yourself in Python. I said 6, he said why?

Then in Sql, I said 8
in Stat & ML, I said 9

Then he started asking ques on stat & ML

What is statistics?

20 mins discussion was on the order of things we use in statistics. Like 1st define Null hyp, define significance level, define effect size, decide sample size, methods of collecting the sample, find test statistics, comparing with population

If you have whole population, do you need statistics?

What is P-value, be crisp on definition.. don't mention to forgot in last, under the null hypothesis is true.

Ques on confusion matrix.

The interview went till 45 minutes.

TIPS- Prepare case studies especially related to Wallet Loading, Customer Retention, etc.

MANIKYAM

Round 1

1. *Error Metrics:*

- Derive the equation for total error in a machine learning model.

2. *Ridge and Lasso:*

- When would you use Ridge vs. Lasso regression? Explain the differences mathematically.

3. *Case Study:*

- Design customized deposit tiles for customers to increase the amount they add to their accounts.
- What data would you use, and which features would be most relevant?

Round 2

1. *Why This Company:*

- Why do you want to work at our company?

2. *Case Study:*

- How would you design a personalized recommendation system to suggest the top 6 games for each customer?

- What features would you use, and how would you measure the success of your recommendations?

Rahul Kumar

Round 1

1. Fundamentals of ML Models:
 - Metrics: When to use, why, and what they mean.
 - When to use Precision and Recall.
2. Data Science Case:
 - Working with non-labeled data.
 - Why was clustering used?
3. XGBoost:
 - How is it different from LightGBM?

Round 2

1. In-depth discussion of everything listed in the resume.
2. Hard Statistics Questions.

Ishan

Round 1

1. **Introduction:** General introduction and background.
2. **Case Studies:**
 - Discussed the approach for case studies from the online test (most critical part).
3. **Work Experience:**
 - Demand Planning:
 - Metrics tracked (STAR approach).
 - Statistical questions on right-skewed data.
 - Define Weighted MAPE.
 - Forecasting goods units.

Round 2 (Leadership Focus)

1. **Behavioral Questions:**
 - Most proud and embarrassing moments.
 - Interests:
 - Research-based vs. Business-based.
 - New technology vs. Tried-and-tested technology.

Aniket Sinha

Case Study:

- Problem Given: Create features and a model within 10–15 mins.
- Inference: Mann-Whitney Test.

SQL:

- Query related to join statements.

SANGITA BHAKAT

1. Written test : the test was rigorous in terms of questions . It has subjective questions where we had to read the case and we had to write the approach and statistical test that we will do etc

Can't remember exactly

After the test 12 to 13 people were shortlisted for the interview.

Round 1 : here they did not give an opportunity to choose a project . Interviewer started from top of resume

1. What is OLS?
2. what else other than OLS ?
3. what is regularisation ?
4. what is multicollinearity and when it is problematic ?
5. how to remove multicollinearity ?
6. VIF what is it , how it works ?
7. metrics used in classification ?
8. What is recall , precision ? Not formula intuition of it
9. Asked about basic statistical tests
10. xgboost mathematical forms
11. hyper parameters and how it is related with regularisation, basically XGboost math derivation
12. L1 and L2 differences ?
13. How boosting and bagging works and differences ? When to prefer what ?

NITISH KUMAR

1. First Round:

- Open to all, test-based. The test was case-focused, specifically designed for gaming companies. You can get sample questions from your seniors (Batch 9).
- Advice: Use real-life examples in your answers for better traction. Very few candidates were shortlisted.

2. Final Interview Rounds:

- First Round:
 - Interviewed by a Data Scientist.
 - Focus Areas: Detailed explanation of ML algorithms and their application in consumer retention strategies.
- Second Round:
 - I was the last candidate of the day, and it seemed they were in a hurry.
 - Given a case study: "How to reduce customer churn?" They expected answers aligned with specific expectations, while I followed frameworks from IIM ABC case books.
 - Also included simulation-based questions in a rapid-fire format.
 - Outcome: Did not make it to the final list.

Nisarg Tole

Round 1:

- Long discussion on regression.
- Core work experience discussion—predicting traffic in a metro line between two regions.
- What metrics would you use to estimate occupancy?
- How would you design headway between two metro cars?
- Why ANOVA in regression
- Regression assumptions
- High leverage point vs. Outlier? What is the impact?
- MAPE is good, but diagnostics are poor—what would you do?
- Regression errors—Equation of error, computation, and impact from a business perspective.
- What is Euler's number, and how is it used?
- What is BLUE?

Round 2:

- How have you used data to extract business insights? Data parking and healthcare data
- How would you approach Churn Analysis it for a business? What metrics would you use?
- Customer retention strategies—How would you identify persuadable customers?

DE Shaw FinRes

ASHUTOSH SHARMA

Round 1 (10-12 mins) – 2 Panelists

Questions from CV only

Some Basic questions on LLM. How temperature affects the performance of LLMs

What is p-value

Case – Client has some data where names of the organization and country are given. There is another dataset where United states of America was written as USA and Republic of China as China. How will you match them?

Tip – Be thorough with atleast 2-3 projects

ARVIND

Case Round (30 groups) –

Newyork Taxi forecasting – How many rides per day?

(Metric – MAPE score) – Approach is important – Interpretability is important (Glass box model)

DE Shaw expects or asks more statistical questions

Regression & Time series is Important

(Around 15 groups were shortlisted for Presentation round)

Presentation Round

If any other model used, How could it have worked?

Round 1 (Same panelists of PPT Round)

TMAY

Why food technology?

Why PGDBA after Food technology?

I had a K-Means project, What is k-means?

If you run the code again, will you get the same cluster or different one?

What is Random forest? Bagging, Bootstrap?

Pseudo code for Bootstrap (with replacement)

Time series concepts – What is co-integration?

VARMAX? Given Target variable, which feature will affect the Target variable?

Regression project – Discussed all keywords written in it? – Assumptions, Techniques used, case or Problem of this project. Why you took that problem?

1 Puzzle based question

10-12 members were shortlisted for next round

SAHIL CHADHA

Case submission round (DCL - DE Shaw champions league)

Case was on taxi demand forecasting. They gave us a dataset, and then asked us to make a forecasting model. There were different components of evaluation criteria: model - interpretability and explainability; general insights from data sets, insights from models, forecasting monetization (how to use this forecast for generating business value or in investment decisions) and business insights. We tried various approaches - from SARIMAX to decision trees. Flow in ppt is a must. Don't over focus on the MAPE score, it's just one component of evaluation. It was a team event - a group of 2.

Case presentation round

Around 18 teams were shortlisted for this round (36 people). It was for about 20 min (though for some students it went for 40 min). 10 min - presentation, 10 min - QnA. questions were asked from model, exogenous variables used and business insights.

Pre round (online)

Around 23 people got shortlisted for the pre round. This was an interview round. the 1st introduction was asked. Then a case was given. We have to open a hotel and we have 2 locations, say Paris and Tokyo. What factors will you consider? The discussion went on demand factors. Then they asked what could be the leading factors of demand for hotels. (It is recommended to go through cases of DE Shaw past year interview experience. They have 3-4 frameworks on which they ask).

Then they asked me to explain the

Regression - Lasso and ridge was asked. PCA advantages. Lasso - scaling required or not.

2 puzzles - angle and tournament problem.

What is the difference between prediction and confidence interval, which one is bigger.

Given 2 time series, what is wrong in calculating pearson correlation factor. (I gave answer related to cointegration - they might have common trend)

Round 1 (offline) (2 members, both PGDBA seniors)

Case - open restaurant at a location, what factors you will consider - lots of cross questions on that.

Explain stationarity in layman terms. Why stationarity as an assumption is required for time series models.

What is cointegration?

Case - based on spell checker. You have to make a model which can estimate the number of incorrect words in a document. You are given 3 scenarios, 1st - you have to submit a model by tomorrow, 2nd - you have one month, and 3rd it's a semester-long project. In all 3 cases, what will be your approach?

Puzzle - simple probability based puzzle. Given a 5 match series between India and Pakistan, where 1st match was won by India, second by Pak, what's the probability that the final will happen on day 5.

Any questions for us.

Round 2 (offline) (2 members, one Ma'am - PGDBA senior)

Introduction ; Went in depth into my time series project (wheat price). There, I have used the USD index as an exogenous variable. He asked why only this as an exogenous variable and how demand and supply dynamics can explain it as an exogenous variable. Lots of cross questioning happened on that and I was grilled on this.

The other panellist asked a tournament puzzle. It was like - there are 3 friends playing badminton friendly back to back matches. The one who gets defeated is replaced by the 3rd one, and one who wins continues to play. This way, 1st one has won 13 matches, 2nd one 7 and 3rd one 14. (not sure of the numbers). Then this was asked - who lost the 6th match - something like this.

Even though it was not tough, it took me time to figure it out. Trick was to 1st find out number of matches played, then move ahead.

Then they asked what the data of Trilytics was about, since I mentioned it in my introduction (that I am part of the team).

Any questions for us.

Soham

Round 1: Case Presentation (~30 min)

Case: NYC Hourly Cab Demand Index Forecasting

- **Approach:** Used SARIMAX (aggregation) + Random Forest Regression (disaggregation) to forecast demand indices with dual seasonality (daily and weekly). Exogenous variables included subway ridership, precipitation, and temperature.
1. What kind of variable is 'hour of the day' (nominal categorical)? What about the demand index (discrete/count data)?
 2. How would you statistically measure the association between a nominal variable and a count variable?
 3. Why did you aggregate data daily instead of weekly?
 4. How would you incorporate subway ridership into a forecast when future data isn't available?
 5. Would lag 2 for subway ridership show a similar pattern? Why or why not?

Difficulty Level: Moderate

Round 2: Online (~30 min)

1. What exogenous variables would you use for your petrol price prediction project?

2. Why did US fuel prices dip in 2020? How would you handle Black Swan events? Can all such events be handled similarly?
3. In your Customer Uplift Modelling project, why did you undersample instead of oversample?
4. Other than Random Under Sampler, what undersampling techniques could preserve data characteristics?
5. Discuss your Power BI project:
 - Did you use DAX?
 - Commonly used DAX expressions?
 - Data integration and transformations?
6. Types of missing data and handling MNAR missing data.
7. **Puzzle:** 17 matchsticks game where the last picker loses. Strategy to win?

Difficulty Level: Moderate

Round 3: Offline (~1 Hour)

1. How do you handle multicollinearity? Is it always a problem?
2. How does an SVM RBF kernel work?
3. What is PCA's underlying concept? Is feature scaling necessary before PCA?
4. Similarities and differences between PCA and SVM.
5. For a refinery with multiple machines, what data would you need to predict daily production?
6. For defective products, how would you pinpoint issues and prioritize categories? Incorporate time-to-failure analysis. Which distribution might model time-to-failure?
7. Is Pearson's correlation coefficient appropriate for time series data? Why/why not?
8. If the correlation between X and Y is 0.8, what is R^2 ? (Answer: 0.64).
9. If X_1 and X_2 have correlations of 0.8 and 0.6 with Y, how would you calculate the combined R^2 ? Are these correlations sufficient?
10. How would you build a spellchecker with a corpus of 10,000 emails?
 - Is it supervised or unsupervised?
 - Training data and labels?
 - Validation metric?
 - Differentiate between corpora of Twitter content and books for this task.

Difficulty Level: Moderate to Hard

Round 4: Offline (~20 min)

1. A student's Chemistry score is missing, with a pattern that Chemistry is the average of Physics and Mathematics. How would you impute the missing score? Would you use row-wise or column-wise imputation?
2. A parabolic data scatter is shown. How would you fit a linear regression? Other than transformations?
3. In the NYC cab demand case, why does precipitation increase cab demand?
4. **Puzzle:** 100 doors toggle game. After 100 iterations, which doors remain open?

Difficulty Level: Easy to Moderate

Surya Kiran

Case Study Round

- Focused on a time series problem with double seasonality.
- Questions were centered around the presentation of the case study.

Result: Qualified for the next round.

Pre-Interview Questions

1. Discussed the time series project:
 - Why specific exogenous variables were selected.
 - Criteria for variable selection (asked by two-thirds of the interviewers).
2. What is negative sampling in Word2Vec?
3. What did you do in the topic modeling project?
4. What is RNN, and how is it different from Word2Vec?
5. Case-based question: How would you build a dashboard for a shipping company?

Ishan

Round 1

1. **Introduction:** General introduction and work experience discussion.
2. **Resume-Based Questions:**
 - Classification and uplift projects.
3. **Puzzles:**
 - Matchstick Puzzle:
 - Incorrect question; required starting second to win.
 - Functions Puzzle:
 - A monotonically increasing function on natural numbers ($x_1 > x_2$, $f(x_1) > f(x_2)$), with $f(f(x)) = 3x$.
 - Solution: $f(13) = ?$.

Aniket Sinha

Case Study:

- Problem: Person can invest in platform A or B. Evaluate and recommend.
- Regression Models: OLS, Lasso, Ridge, Adjusted R^2 .
- Discussed data source, project goals, and column removal strategies.

Peehoo Jain

Pre-Round

1. **Case-Based Questions:**
 - Exogenous variables: What additional ones could be used?
 - Weekly trip counts: How would you use this data?
2. **Puzzles:**
 - 25 horses: Determine the fastest horses.
 - Puzzle on clustering and handling outliers.
 - Dartboard clustering: How to cluster points forming concentric circles.
3. **Machine Learning Questions:**
 - Concepts of clustering and classification.
4. **Time Series Questions:**
 - Identify and differentiate time series using ACF plots.
 - Types of time series equations.

Round 1 Interviews (40–60 mins, 2 Panelists)

1. **Case-Based Questions:**
 - Simple scenarios: What would you do?
 - 18-digit number puzzle: When the 18th digit is moved to the first position, the number doubles. Logic?
 - Graph: How would you fit a model to the data?
 - Oil prices: Identify potential exogenous variables.
2. **Candidate-Initiated Questions:**
 - Skill gaps in students.
 - How much time is given to understand the business?
 - Addressing financial skill gaps for non-finance backgrounds.

Round 2

1. **Puzzles (from Geeks for Geeks):**
 - 25 sticks: You can pick 1, 2, or 3 sticks. Winning strategy?
2. **Graphs and Cases:**
 - Draw graphs and explain how to model them.
 - Order determination of time series. (*Recommended: IIT KGP RTSM for self-study on time series core concepts*).
3. **Miscellaneous:**
 - SPA time series project: No questions asked.

Manas

Round 2 (Direct Selection)

1. Questions:

- Why not IR (Interest Rate Modeling)?
- Personal introduction and passion for math.
- Puzzle (Geeks for Geeks).
- Time series basics.
- No case studies in this round.

SANGITA BHAKAT

1. They rolled out one case comp based on a time series forecast of the number of taxis in New York .

After submission of the case they had given a short list for 1st round.

1st round : pick one project
I chose regression.

Then the same basic questions on regression starting from project brief to why regularisation , benefit of it ?

What is multicollinearity and how to mitigate that ?

When multicollinearity is important ?

What is VIF?

questions on statistical tests:- t-test and f-test and how they are related.

There were 3 panelists.
One asked the above questions.

Then the second one asked about the probability , and gave one puzzle based on probability . Can't remember now

3rd one asked a case study

Case was , if Deshaw wants to expand in a new city in India , which location should it choose and why ?

What are the factors to consider , what are the things to Predict etc etc

2nd round : they started with time series project and asked different methods that we applied? Why of everything that we have done in the project ?

One question on probability which was the same for all candidates.

One question on spurious regression ?

NITISH KUMAR

1. Case Competition:

The first stage was a case competition (group of 2), open to all. The task was to perform forecasting on a given small dataset using a variety of exogenous variables. The evaluation was based on both the MAPE score and the simplicity of the algorithms used.

- Our approach: We achieved the best MAPE score using a combination of XGBoost and ProphetB.

- Advice: Involve as many relevant exogenous variables as possible. Use web scraping and free data sources smartly. Experiment extensively but keep the algorithms simple.

2. PPT Round:

Shortlisted teams from the case competition presented their thought process through a PowerPoint presentation.

- Our strategy: We highlighted the exogenous variables we considered, our integration methods, and the rationale behind our approach. This was well-received.

- Advice: Be thorough in explaining your methodology and focus on the reasoning behind your decisions.

3. Preliminary Technical Round:

This round was conducted by three PGDBA seniors who evaluated our technical skills and foundational knowledge.

- Topics: Machine Learning algorithms and the mathematics behind them.

- Advice: Go through ISLR thoroughly to strengthen your understanding.

4. Final Interview Day:

First Round:

- Duration: ~1 hour

- Key Insights:

- There seemed to be a bias toward IR profiles, and I didn't have prior relevant experience. If you are outside their "hotlist," you'll need to put in extra effort, and your interviews might be more challenging.

- Focus Areas:

- Puzzles

- Probability questions

- Time series concepts (e.g., why regression cannot be directly applied to time series data).

- Basic finance knowledge (helpful but not mandatory).
- Advice: Study time series in detail if De Shaw is your dream company.

Second Round:

- Duration: ~45 minutes
- Key Questions:
 - If you randomly mix time series data, can you apply regression to it? Why or why not?
 - What geopolitical factors would you consider when modeling the energy market?
- Advice: Be well-versed in regression and time series concepts, and understand their real-world applications.

SAMEER GUPTA

Case Competition

Objective:

Predict the demand for taxis in the USA using a provided univariate dataset.

Evaluation was based on the Mean Absolute Percentage Error (MAPE).

Duration:

The competition lasted two weeks, with a leaderboard updated based on model performance.

Solution Approach:

Used a combination of XGBoost and SARIMA models.

Final rank was around 6 or 7 on the leaderboard.

Case Presentation Round:

Key Focus:

A lot of emphasis was placed on the exogenous variables used in the model.

Candidates needed to be clear, thorough, and confident about their choice of exogenous variables.

Preparation Tip:

Go through IIM Calcutta's SAAPM material (3rd semester) to strengthen knowledge of time series and exogenous variables.

Insight:

DE Shaw places significant importance on the use of exogenous variables in time series models.

Pre-selection Round Interview

Questions Asked:

Time Series:

Assumptions of time series models.

Selection criteria for exogenous variables.

Methods to analyze two time series together (e.g., Granger's Causality Test or Cross-correlation).

Finance (CFA):

Comparison of companies: identifying growth vs. value stocks.

Linear Regression:

Assumptions of linear regression.

Key Advice:

If you don't know the answer, admit it. It's better to be straightforward than to mislead the interviewer or yourself.

Final Round (Placement Day)

After clearing the pre-selection round, the final round of interviews took place during the placement process.

Questions Asked:

1. Exogenous Variables & Time Series:

Focus on the choice of exogenous variables.

Assumptions related to time series models.

Application of knowledge to real-world scenarios, e.g., predicting oil prices (since your previous work was related to oil prices).

2. Work EX:

Questions about your work experience (Work EX), especially related to how you would apply your knowledge to specific tasks like predicting oil prices.

3. Probability Question (Logical Puzzle):

The question involved three drivers (A, B, and C) who play a series of matches. The sequence of matches is based on the outcomes, with specific conditions like:

A plays against B, and B loses, so A plays against C, and so on.

You are tasked with predicting who plays against whom on a specific day after several rounds.

The question was logical and required a systematic approach to solve, with the interviewer guiding you when you got stuck.

Key Takeaways & Advice:

Superficial Level:

Most questions in the final round seemed to be a repeat of the ones asked in previous rounds, especially focusing on exogenous variables and time series.

It's important to be prepared for both theoretical questions and practical applications, like predicting real-world variables (e.g., oil prices).

Practice Puzzles & Logical Questions:

GeeksforGeeks is a good platform for practicing puzzles and logical questions, as these are common in such interviews.

Helpful Interviewers: If you get stuck, interviewers tend to help you and guide you toward the solution, so it's important to stay calm and approach the problem logically.

Key Takeaway:

For DE Shaw, unless you are an IR (Institute Rank holder) or someone who is exceptionally intelligent, extremely thorough with time series analysis, and able to demonstrate a deep understanding of the subject, it might be better not to invest too much time and hope in getting selected. DE Shaw has a highly selective process, and they are particularly keen on candidates with a strong grasp of time series concepts, especially when it comes to selecting exogenous variables and handling complex models.

While the process may seem approachable, it's important to recognize that the competition is intense, and their selection often leans toward those with a deep, niche understanding of the domain.

Varun Bhagwani

DeShaw:

Contest: Forecasting and business understanding

Round 1: Online Interview

Introduction.

Since I mentioned my interest in Football, the next question was how to make clusters of goals scored during a tournament in different parts of the goalpost. Some followup questions.

I answered DBScan and explained. He then wanted me to use k-means clustering. Then some questions on Kernels.

Discussion on my project on time series forecasting on Natural Gas. Follow up questions on SARIMAX. Is 272 data points enough for a time series forecasting?

Abhishek Anand

1. Introduce yourself?
2. Which project do you wanna talk about?
3. How did you tune the hyperparameters of isolation forest?
4. Why did you use Logistic Regression & Random Forest? Isn't RF immune to outliers?
5. Did you examine your model's false positives to understand what kinds of patients?
6. In your uplift project, why did you use a random undersampler? What's the issue with undersampling? What other ways of undersampling can be used here?
7. You have a MAPE of 20%. Is that good? How do you interpret these results?
8. How exactly did you optimize portfolio allocation? Explain Sharpe Ratio in simple terms.
9. Matchsticks puzzle.

Madhur Rawat

Pre-Round: Technical Screening

- Time Series & Regression: General questions.
- Bayesian Methods: One basic question (since it was a subject I studied).
- Puzzle:
 - In a tournament where the better-seeded team always wins, what is the probability that the 1st and 2nd seed meet in the finals?
- Hypothesis Testing Scenario: Given a situation, determine:
 - Null & alternate hypothesis.
 - Test to use.
 - Underlying distribution.

Round 1: Case Studies

- A company builds office spaces and leases them to multiple businesses.
 - Analyze risk and profitability.
 - Several follow-up questions on the same.
- Given a dartboard with points plotted, design a clustering algorithm where points closer to the center are in one cluster and points farther away are in another.
- Given a list of company names, reduce them to count unique companies (e.g., *Reliance* and *Reliance Pvt Ltd* should count as one).

- Given large-scale transaction data, answer: How to monetize the data? How can Shaw use it for insights and revenue?

Round 2

- Given 9 horses and 3 tracks, determine the top 3 fastest horses with the minimum number of races.
- Given 20 years of daily weather data, predict for the next year
- Given a curve that looks like $y = e^x$, model it in the simplest way possible.

Namit Pulyani

Pre-Round:

- Questions on OLS, Ridge, and Lasso regression.
- What is Out-of-Bag (OOB) error in bagging?
- Basic t-test question.
- One mathematical puzzle

Round 1:

- Given a plot of all goals scored in penalties by an opposition team (dots plotted inside a rectangle representing the goal area), how would you analyze the data?
- Slightly modified time series questions.

Round 2:

- Given daily booking data for a year, and a point estimate for the next month's total bookings, how would you estimate the number of taxis booked for each day of the month?
- You have 25 horses and 5 race tracks. How will you determine the top 3 fastest horses
- Further questions on time series with slight variations.

DE Shaw FinOps

SAURABH

Brief intro?

Asked about one project of your choice, why this model why that model, why r^2 was high, why do you think XG Boost works, pls remember the features

What is decision tree what does XG Boost really do, what is information gain, what is entropy and how is splitting exactly done based on info gain and entropy?

how XG Boost differs from decision tree?

what we see when we split give the formula

Focus on one project of your choice

Reason for XG Boost - I have used in regression

AKSHAY K

Round 1

Talk to me about the project you did – Free lancing project (Image classification project – liveliness checking – KYC verification)

Questions on what model is used?

Tip – Pick a project that you are confident with.

Case – How many invoices are likely to be in future months (Time series)

Asked about Approach?

How would you check a time series project in production?

Round 2 (All cases)

Case question about how you would classify alerts related to insider trading as being of Low, Moderate or High importance. Historical data for all trades and whether there was an alert (but no classification) associated with them is available.

This would be an unsupervised learning problem.

HR questions – Why did you choose FinOps role not FinRes?

How would you see yourself in 2-3 years? What kind of problems you expect to be working on after 2-3 years?

AKSHAT MISHRA

Pre-Round (10-15 mins) – Online

TMAY

Pick one favorite project from CV? (I picked Regression project) - Some Cross questions on that.

Why PCA is performed? What is PCA? (Questions framed from the project statements and keywords written)

Data science case study – Usecase was from NLP

There are several news coming from trading, geo-political scenario. You have to go through all the articles for trading purpose. In those articles too, some were small and some were quite long. We have to devise some technique that could trade 1 stock based on that news. Let's say for example, If there are several articles, we have to trade wipro share. How would you do that? You have 50K articles to go through. You have to trade a stock based on the articles.

(Ans – I said that I will filter based on Company and recency of the article. More recent the article was, more impact on the share price. I mentioned we can summarize the articles by using NLP techniques like Stemming and Lemmatization to reduce the length of articles and summaries using GenAI. Based on the summarization, we can trade the share.)

Tip – Practice of Data science cases will be helpful for both Pre and final round.

Final Round – Case based round (On campus) (30 mins)

Case study was based on Use case of Gen AI -

There is a system to detect the chances of insider trading in DE Shaw. So for each trade, they can identify it as risky trade or non-risky trade. Risky means that it has some chances of insider trading but other don't. Out of the trades that their system is categorizing as risky, they wanted me to rank the trades based on their risk. So, for ranking these transactions, they asked me to first gather some features based on which you can rank them. After that, they asked me to formulate an algorithm with which you can do this ranking of transactions. (It was quite a lengthy discussion around 30 mins)

KAPIL BHANWARIWAL

DE Shaw – FinOps (Pre-Round) – 15-20 mins

TMAY

Asked about my favorite project – (Told about ML project – Predicting Taxi fare using Neural networks – Developed code from scratch)

Why did you use Neural Networks instead of other model?

Ans – Because this was academic obligation

If you have to do this project again, How are you going to do it?

Ans – I will consider 3 models – Linear regression, Tree based, Neural networks

Which one will you finally choose?

Ans – Reject Neural Networks (no explainability) and Linear regression (Not able to capture non-linearity)

Case study – (5 – 10 mins)

Live Problem in their Operations Dept - We receive a lot of bills from vendor, how will you convert this data into structured format? (Into tables)

My /Ans – Columns can be Name of Vendor, Amount paid, Due Date. I said 2 approaches can be used to solve this problem. 1) Image recognition (OCR) 2) something related to NLP. They asked me to think in the lines of NLP techniques. I am not sure; the correct answer may be Named entity recognition or RAG's can be used.

ASWIN

Round 1

TMAY

Why you took PGDBA?

What was your previous role as a Business analyst?

Questions on projects (If we have a finance project, interviewers want to focus on that)

Regression project – Based on CAPM (Financial analysis)

NLP Project – What is embedding? Why do you vectorize it? How can you leverage this project in Finance domain?

RAHUL SINGH

Screening – CV based Shortlisting

PREROUND

Pick Any 1 fav project.. chose Time series, we haven't used exogenous variable in it, he asked what exogenous variable you can think off, if there is sudden covid like situation, is your prediction be valid.

Case Study on you have thousands of invoices in pdf form, you have to use data analytics to properly extract the info from it.

Tips- Be thorough with Time Series & Regression Project, be ready for cross questions.

Vineet

Round 1: Online

- **Introduction:**
 - Tell me about yourself.
 - Favorite project: Why did you choose it?
- **Technical Questions:**
 - Benefits, datasets, and feature engineering-related queries.
 - Working of the models used in the project.
 - Feature importance in Random Forest.
 - Criteria for selecting evaluation metrics.
- **Case Study on NLP and Finance:**

- Given a dataset of company names (legal names), how would you map them to their usage names?
 - Example: *Apple Inclusive* (legal name) → *Apple* (usage name).
- How to link news articles to companies (map articles to the correct company).

Round 2: Offline

- **Technical Questions:**
 - Clustering concepts, including silhouette score.
 - Scenario-based question:
 - Given a dataset containing `timestamp`, `price`, and `instrument name`, how will you identify suboptimal fields?
- **General Discussion:**
 - Company and role-related queries.
 - Discussion on ongoing projects related to Power BI, Tableau, NLP, and LLM.

Shambhawi

Case Study Questions:

- Questions related to the NLP case study.

Data Analysis:

- High-volume trading data discussion.
- Given opening and closing prices, calculate the volume traded.
- Address cases where opening and closing data are faulty (no true labels available).
How to handle this?

Quantifying News:

- Discussed approaches to quantify the impact of news on trading.

Awadhesh Mahato :

Resume Shortlisting

Intro

Favourite Project: Explain, Questions related to terms mentioned

Chose Classification project

Oversampling vs Undersampling

Random forest vs XGB

How feature importance is calculated?

Which were top 3 features?

Case: Forecast daily invoices volume for next month
Which exogenous variable would you take?
Which evaluation metric would you consider?
How would you make your model more robust?

PARTH PANDEY

DEShaw (Financial Operations):

CV Based Shortlisting

Round 1:

Favourite Project - Regression- Data related questions - Polynomial Regression - Lasso/Ridge Regression - XGBoost (for improvement).

Suppose you have been given two excel files of companies names with many overlaps, for example, Apple in 1st file and Apple Inc. in 2nd file. How will you combine these two to make a single file without losing any name?

Round 2: (Online mode)

Company has a black box model which has trading data. This particular model raises alarm on that particular trading where it thinks that there is something phishy (maybe not followed compliance or used internal information). How will you sort the list of these types of problems/warnings? What factors will you consider to be the most important?

(TIP: Ask as many follow up questions as possible.)

Himanshu Singh

First Round: Case-Based

- Problem: Thousands of bills are manually parsed and entered into a system. Devising an automated solution.

Second Round: Case-Based

- Problem: Compliance department receives numerous insider trading alerts daily but has never found a true positive in ten years. Need to design a system to assign priority levels to alerts.

Nisarg Tole

Case Study & Presentation Round (Time Series)

- Seasonalities, exogenous variables, multiple models.
- Which exogenous variables were used?
- Business impact of predictions?
- Optimized code submission on their website.

Technical Interview

Round 1:

- Tell me about yourself.
- Why do you want to join DE Shaw?
- Why did you use this model in your case study?
- Discussion on a trading case competition.
- Detailed discussion on Time Series project, exogenous variables, and forecasting energy requirements.
- Multiple seasonality—what approach would you take?
- Does pure Moving Average (MA) exist?
- Co-integration vs. Correlation?
- Types of ADF Tests.
- Statistical test for normality?
- ANOVA concepts.
- Trading project discussion:
 - How was the strategy implemented?
 - When to take exit decisions?
 - How to identify if the strategy is failing?
- Probability puzzle: 4 locks and 4 keys.

- Case study:
 - Given parking & healthcare data, how would you use it for financial gain?
 - Can you use clustering to determine the number of goals scored?
 - What model would you use?
 - Can this be solved with basic regression?

Oliver Wyman

KAPIL BHANWARIWAL

Round 1 (20 mins)

Introduction

What is Bagging? What is Boosting? What is Generalization error? What is Bias – Variance trade off?

When will you use Bagging? When will you use Boosting?

Ans – Bagging is used When the objective is to use reduce variance, Boosting is used when the objective is to reduce Bias. This is a very theoretical answer and it depends on the situation and gave a detailed answer.

They gave me a scenario – If you need to predict a default which is a binary variable whether a customer will default or not? Which models you are going to use and why?

Ans – 3 models – Logistic regression (Linear models), Tree based models, Neural Networks

Which one will you use and why?

Ans - I answered I will reject the Neural networks because it is not explainable and also reject the logistic regression for its linearity. So, I will go ahead with the Tree Based models.

Have you worked on Light GBM or any gradient boosting machines?

From HR,

If you are given a situation to handle conflict, how do you handle it?

Ans – I clarified from them who are Stake holders involved and about the situation in detail, then I said that I will first talk with all the stake holders involved, ask their opinions and based on that I will take the decision which is for the greater good of the organization.

Round 2 (Online round – Director + Sr. Data Scientist)

Introduction

Tell me your favorite project

Ans – Taxi Fare prediction project using Neural Networks

Explained in detail about the project for 3-4 mins.

Most of the questions were on the line of Business perspective or value

Sr. Data Scientist questions:

One question from my Regression project – Used car price prediction – What is multi-collinearity and problems associated with it? How do you figure it if there is multi-collinearity?

One question from Countries clustering project - You used 2 algorithms, K-means and C-linkage? Suppose if you must use only 1 algorithm, which one will you use and why?

Ans – (If you have a large dataset, C-linkage is going to take a lot of time because it's time complexity is much higher than K-means. Even when I did the project, I observed C-linkage took around 15 mins to cluster just 167 data points. So, I will use K-means.

What is Jaccard similarity?

Ans – I used it to compare the results of K-means and C-linkage algorithms.

RAHUL SINGH

Screening – CV based Shortlisting

Test

Interview-

Choose your fav project, I chose CNN.. cross ques on that

He picked CNN project from my resume & asked the various ques on finding the dimensions of output matrix based on some architecture.

Keep clear these things, these are the most common asked ques from CNN.

Another ques could be how image is converted into matrix form.

Soham

Round 1 (~15–20 min)

1. Why do you want to work as a consultant?
2. Discuss one good and one bad thing about your EXL projects.
3. How would you handle a difficult client unwilling to listen?
4. Assumptions of Linear Regression?
5. What is multicollinearity? How to handle it? Impact on coefficients?
6. Explain the structure of LSTM. What is ROUGE-L?
7. Difference between Encoder-Decoder and LSTM. Advantages of each?

HARSH RAJ

(20-25 Minutes)

1. After introduction, the interviewer asked me to explain about anything present in resume apart from the projects. He asked me about work experience, interests.
2. The panel consisted of two people, one was asking HR questions - How would you resolve conflict?
3. Why Oliver Wyman
4. He asked questions from my classification project - F1 score, AUC ROC
5. Bagging vs Boosting

Subhav Chauhan

Pre Round-

One SQL question (JOIN) and 3 python questions (one was on Matrix Multiplication).
MCQs - On Pandas , ML and basic python programming.

1st Round -

(Selection Criteria- Pre Round + CV)

Introduction & Why this course?

Discussion on my project on Propensity Modeling. First, they asked me how it differs from Predictive Analytics, followed by in-depth questions on data cleaning and preparation. Lastly, we discussed the XGBoost algorithm, including the types of data for which XGBoost is most suitable.

PARTH PANDEY

Test:

2 questions on Python, 2 on SQL and around 10 MCQs on ML/ Statistics.

Round 1:

Favourite Project - Regression- Data related questions - Polynomial Regression - Lasso/Ridge Regression - XGBoost (for improvement).
Detailed discussion on XGBoost.

Why do you want to join Oliver Wyman?

Why consulting?

Round 2:

Tell me about yourself.

Project- Regression - Same discussion as above.

When to use XGBoost and Random Forest. What are the differences?

Himanshu Singh

First Round:

- HR Questions:
 - Introduction.
 - Why consulting?
 - How would you handle conflict? (Team and Client)
 - Several follow-up questions based on responses.
- Tech Questions:
 - Resume-based questions, focusing on bold words.
 - Clarifications on why multiple techniques were used in a single project.

Second Round:

- Asked to discuss a project with multiple counter-questions.
- Questioned why ML was used instead of LLMs.
- LLM disadvantages as compared to what you have done.
- Asked how I would change my approach if I were to redo the project now.
- Discussion on PCA, LDA, and other techniques in the resume.
- What is PCA and why? how would you describe it mathematically?

Mastercard AIG

ASHUTOSH SHARMA

Round 1 - On campus – (15-20 mins) – 1 Panelist

TMAY

Some questions about his work experience on Data engineering

Some simple questions on Precision, Recall, p-value

Describe one of your projects

(Ans – Explained the project on Uplift modelling. There were some cross questions on it)

What is Uplift? How it is different from regular classification problem? Is it a supervised or unsupervised problem?

If you have a dataset with large number of rows but with a smaller number of features or columns, which ML Algorithm do you start with?

Tips – Be confident about what you have written in your CV. Be thorough with at least one of your projects

Round 2 – (15-20 mins)

TMAY

Some questions on regression – What is confidence interval? What is prediction interval?

Asked about elastic net regression and asked to draw the contours for elastic net regression

How it is different from Ridge and Lasso regression?

Questions were asked on my Deep learning project on Mechanism of action of Drugs.

Situation based question – Let's say we are creating a feature in Fitbit app to predict heart attack. What are considerations you will consider from end-to-end model building to deployment?

Which is more important in this case – Precision or Recall? (Ans - I said Recall)

If you falsely detect a person has a heart attack, it will be costly? (Ans -We should check Fbeta score giving more importance to recall, I will also focus on Latency and throughput)

How do you check if your model is biased?

How will you productionise the model was main crux of this interview

PRABAL

Mastercard AI Garage (1 Panelist) – 20-25 mins

TMAY

What is your favorite project? (Ans – Regression)

Explain the Regression project?

Why don't we use LSTM for image processing? (Bcoz I had a project on image captioning (It has both CNN and LSTM part))

(Ans – Basically we don't have a sequential data, and the spatial pattern of the image data also cannot be captured in an LSTM)

Do you have any questions? (Ans – I asked a question from their PPT)

KUNAL

Round 1

TMAY

VGG- 16 vs RESNET – 50 (CNN models) – Difference between them other than residual connections in RESNET – 50?

(He was interested in last layer difference in Neural networks)

Explain one of your projects? I explained NLP project

What is AUC score you mentioned in classification project? Do you know Precision recall curve? How does it look like?

What is pooling and stride in CNN?

Round 2

Mostly on my classification project and Regression project

What is difference between Precision and Recall? When to use which metric?

What is multi-collinearity? What is Ridge($\lambda \rightarrow 0$) and Lasso regression($\lambda = 0$)? Why the λ values are like this in the techniques?

Case study – Hospital Data – What will you use? Precision, Recall or F1 score for this data?

SUBHAJIT TARAFDAR

2 cases before interview – randomly selected people

Video (3-4 mins)

PPT (Not Mandatory for AIG, Mandatory for ACS)

Cases – 1. Fraud detection in communications – Phone calls etc. – You need to use NLP to detect.

2. Customer retention strategy

Round 1 (15-20 mins) – On campus

Classification project – heart disease classification

Asked to tell this project

What Imputation techniques have you used? (Used smote). What other methods you could have used?

SMOTE was not good. Why it's not good?

Classifier used – About Random Forest, Decision trees?

Working of Random Forest and Decision trees?

Round 2

Master thesis project. Some CNN questions for 2 – 3 mins.

Graph neural network based project –

- How Graph convolution network is different from Graph attention network?
- How they work?

Given a situation – 1 video – multiple photos – Have a sequence. You need to build a classifier to see the sequencing of photos are correct or not?

If you have multiple classes, you are using cross entropy for loss classification. Instead of 0 and 1, different values like 0.3 and 0.7 are assigned. How do you calculate Loss?

AKSHAY K

Talk about 1 project of yours

Questions on that project

Questions on highlighted terms in the resume

MANIKYAM

1. *Regression Project Questions:*

- Can you explain one of your regression projects in detail?
- How does PCA work, and when would you use it?
- What is VIF, and how is it used in feature selection?
- Which metrics did you use to evaluate your model, and why?

2. *AIC and BIC Questions:*

- What are AIC and BIC, and how are they used in model evaluation?
- What is the difference between AIC and BIC, and when would you prefer one over the other?

Raghavendra

Rounds 1 & 2 (Combined)

- Asked to pick a project for discussion (chose an NLP project).
- Project details:
 - Customers' needs based on Amazon reviews.
 - Discussion on metrics used for each case.
- Handling underrepresentation of a class in datasets.
- Graph Neural Network (GNN) questions (based on resume mention).
- Relocation question:
 - Are you okay with shifting to Gurgaon?

Soham

Round 1: Technical (~10–15 min)

1. Difference between AUPRC and AUROC. When to prefer which?
2. Differences between area-under-curve metrics and standalone metrics like precision, recall, and F-score.
3. Draw and explain a sample ROC curve. What does the top-left corner represent?
4. Can precision and recall both be 1? How would you determine the best threshold?
5. What is TPR also known as? Is specificity the same as precision?
6. Statistic summarizing the ROC curve's performance?
7. Discuss your Customer Uplift Modelling project.

Round 2: Technical (~15–20 min)

1. Walk through your AIDS Fatality Detection classification project:
 - Features used?
 - Feature association assessment?
2. Why SMOTE and not undersampling? Can SMOTE handle categorical features?
3. Why report recall for the classification project?
4. Is class imbalance always problematic? Can handling it cause bias?
5. Differences between XGBoost and CatBoost. What are symmetric trees in CatBoost?
6. For a smartwatch detecting heart attacks:
 - Which error type to minimize?
 - Required input variables?
 - Handling trade-offs between error types 1 and 2?

Surya Kiran

Round 1

1. Explain two favorite projects in detail (20 minutes).

Round 2

1. What is an RNN, and how is it different from a transformer?
2. What is Word2Vec, and how is it constructed?
3. Given a highly imbalanced dataset, how would you handle it?
 - Focused on tweaking optimization algorithms rather than using techniques like SMOTE.

Shambhawi

Rounds 1/2

1. **Project Discussion:**
 - Discussed 3-4 projects mentioned in the CV in detail.
 - Created alternate scenarios for the projects and asked how to proceed in those cases.
2. **Metrics:**
 - Precision, Recall, F1 Score: Explained how they vary.
 - How these metrics were applied in the candidate's projects.
3. **Behavioral Questions:**
 - General introduction.
 - Why AI Garage?
 - Do you have any questions?
 - *Advice:* Always ask a question or two. Relate the query to the pre-placement talk (e.g., What makes AIG unique, their work culture, etc.).

Aniket Sinha

Resume Discussion:

- Work experience-focused.
- Stat classifier: Easy questions.

Technical Concepts:

- Meta learners and base learners.
- Random Forest: Feature importance and its calculation in decision trees.

Business Problem:

- Discussed business impact in the classification problem.
- Consulting-oriented, less focus on deep data science concepts.
- Asked about feature engineering for improving the model.

Manas

Round 1 (Easiest Interview)

1. Technical Questions:

- KNN, K-means, DBScan.
- XGBoost vs. Random Forest.
- Assumptions of linear regression.

2. General Advice:

- Focus on explaining algorithms clearly.
- Group study and using resources like StatQuest are highly beneficial.

Round 1 (15 mins)

Panelist: Senior person from Mastercard, not a PGDBA alumnus.

1. Introduction:

- Brief personal and professional introduction.

2. Project Discussion:

- Time Series: Residual analysis, ACF, and PACF plots.
- Clustering Projects: Elbow plots, silhouette score.

3. NLP:

- Converting words to vectors.
- BERT overview and applications.

SANGITA BHAKAT :

CV short list

Round 1 : intro

Explain your classification project .

After explaining, he asked basic questions on the confusion matrix like why recall is important here?

What will you do with imbalanced data ?

What are the feature selection methods applied into it ?Also why ?

Any metric that can be used in case of an imbalance dataset?

Do business and research go hand in hand ?

What is the meta model used for ?

Asked two three basic questions from RAG project of my resume

Why do we use RAG ?

Embedding details ? Like how embeddings created / works etc.

Round 2 : Interviewer asked to choose one project that is not being asked in ist round .

I picked a regression project.

What is regularisation ? Intuition of it ? Then in detail what we had done in the project and why we had done so ?

Basic questions on regression analysis like test to select features etc

Then small discussion on work ex.

Why AIG?

HARSH RAJ

(15 Minutes)

1. Explain any one of your projects.
2. What is R2 score
3. The interviewer then asked me to explain a deep learning project.

4. F1 score vs AUC PR, explain their application
5. Bagging vs Boosting

AWADHESH MAHATO

Resume Shortlisting

Round 1:

Intro

Explain work ex role

Was it only visualisation or something in modeling also?

Explain your CNN project

Asked about the terms mentioned like data augmentation, IoU, dice loss

Round 2:

Intro

Explain work ex role

Explain your CNN project

Asked about the terms mentioned like transfer learning

Why data augmentation?

What's fine tuning?

What is U net?

Explain each clustering algorithms mentioned.

Namit Pulyani

Round 1:

- Discussion on Precision, Recall, and when to use which.
- In-depth questions on ROC and Precision-Recall curves.
- No questions on DL or puzzles.

Round 2:

- Questions related to multi-class classification.
- Questions on RNN, LSTM, and CNN.

Nisarg Tole

Case Study Competition

- Option A: Analyze customer churn for a bank without any given data and propose a solution.
- Option B: Detect SMS-related frauds using Deep Learning.

Round 1:

- Tell about yourself.
- Questions about work experience (Maharashtra Metro)
 - Estimate the number of travelers for a bullet train between Mumbai and Pune.
 - Should you invest in this project?
 - Basic Time Series questions—AIC/BIC.
 - If you have two railway lines, how do you determine which one is more seasonal?
- Discussion on Time Series project, stationary time series.
- How would you rate yourself on the technical front?

Round 2:

- Puzzles: Camel and banana puzzle, 3 bulbs and switch puzzle.
- Coding and SQL questions.
- Case study on regression for a few candidates.

Round 3: HR Round

- Where are you from?
- You know our stipend—what percentage chance that you will join?
- Basic discussion on stipend, perks, and filling personal details.

Soumendu Mandal

1. CV-based Shortlisting
2. Interview Rounds

Round 1:

- Introduction.
- Interviewer picked Customer Requirement Identification project
- Data Collection & Cleaning—how was it done?
- LDA —explanation.
- Coherence Score—what does it signify?
- Why was sentiment classification needed in the project?
- LSTM workings—why used? Did you try other algorithms?

Round 2:

- Time series project discussion.
- Classification project discussion.
- Chat-based RAG (Retrieval-Augmented Generation) chatbot—how to improve it?
- Classification Forest—concepts and applications.

Mastercard ACS

ASWIN

Round 1

Intro

Explain any one project

Ans – I explained about NLP Project. Basic questions from the projects like Why you used SMOTE?

What are the shortcomings of the project

Ans - I didn't have a great accuracy score. Reasons for that are computational limitations and dataset size is small.

Then they started asking about the RTSM project (CAPM) – What is CAPM?

What is FAMA FRENCH? What are the 3 factors and 5-factors models?

Then they asked about timeseries – What is the time series? Give me some example of Time series?

What are your hobbies?

MANIKYAM

1. *Guesstimate Questions:*

- Estimate the number of flights between Visakhapatnam and Kolkata if a new airline is launched. Walk me through your assumptions and approach.

- How would you estimate the weight of an airplane? Discuss your reasoning and process.

AKSHAT MISHRA

Online Interview

Questions were displayed on screen, and you need to answer those and record your video

Have you faced any situation in which you had to multitask either in your student life or professional life?

Case on retail shop – A retail shop is closed. Why is the retail shop closed? What will be the impact of it's closure on other shops of same brand?

Round 1 (15 mins)

TMAY

Based on that, Cross questioning about my profile. How was my work exp?

Questions on the regression project – What are the basic steps that are required before applying the regression?

Guesstimate – My previous work location was Greater Noida. Let's say we have to start the Bus service Aligarh to Noida. You have to estimate How many buses are required based on the people who travel? (This guesstimate basically requires Top Down approach. We have to divide population into Lower income, middle and Higher income category and follow the process. This process is available in all of the casebook. So, such guesstimates can easily be solved)

I used to play TT at state level. Generic question on TT – Have you ever dreamt of being a professional athlete?

Round 2

2 Puzzles from geeks for geeks (There are specific set of puzzles, same question were asked repeatedly to every person)

What is your favorite coding Language? Answered Python. Then asked my favorite library? Answered Pandas. Then asked why everyone has their favorite library as Pandas?

Tip for ACS – Go through CV thoroughly, prepare Puzzles, guesstimates from casebooks.

Hemanth

MasterCard - ACS Role Selection Process

Interview Overview

- **Number of Rounds:** 3 (2 Technical Rounds, 1 HR Round)
- **Panelists:** One panelist per round

Round 1: Technical

Key Topics and Questions

- **Puzzles** (sourced from GeeksforGeeks):
 - *The Camel and Bananas Puzzle:* A classic puzzle often asked in interviews.
 - *The 3 Switches and 3 Bulbs Puzzle:* Determine which switch controls which bulb, entering the room only once.
- **Conceptual Questions:**
 - *Multicollinearity:* Definition and its implications in regression models.
 - *Regression Project:* In-depth discussion on the project, focusing on technical aspects.
 - *Time Series:* Basic questions about time series analysis.
- **Case-Based Question:**
 - "If you are setting up a new shoe store in Kolkata, what are the 3 most important features to consider?"
- **Theoretical Question:**
 - "If all the coefficients of a regression line are zero, what would be the equation?"

Round 2: Technical

Key Topics and Questions

- **Guesstimate:**
 - "If you were to build a hospital in Hyderabad, when would you break even?"
- **Personal and Professional Motivation:**
 - "Why did you leave your role as an Inspector to join PGDBA?"
- **Consulting Skills:**
 - "What skills do you think are essential for a consultant, and how would you assess yourself in terms of these skills?"
- **Regression Project:**
 - Follow-up discussion on the project from the first round.
- **My Question:**
 - "How does MasterCard contribute to Financial Inclusion?"

Rahul Kumar

Round 1: Stress Interview

1. Geeks for Geeks puzzles.
2. Easy SQL questions based on merging datasets.
3. Work experience-related questions.

Round 2

1. Basic ML Questions:
 - Regression and Classification concepts.
2. Guesstimate: Real-world application-based questions.
3. Project-related questions.

Ramakrishnan

Round 1

- **Scenario-Based Question:**
 - Imagine you are an Air India manager. How would you handle promotion and pricing during the festival season (loyalty program)?
 - Features to consider: Age, gender, business, economy.
- **Work Experience Questions:**

- Why did you not transition to the data team in your previous role?
- Business-simulated questions related to work experience.
- **Data-Related Questions:**
 - How would you gather data for the problem?
 - Importance and methodology of conducting a pilot study.

Round 2

- **Puzzles** (common for all candidates):
 - *Camel and Banana Puzzle.*
 - *Rod Puzzle* (Returning the rod in one week).
- **Technical Questions:**
 - SQL: Easy queries.
 - Standard CV questions: Discussed experiences listed in the resume.
 - Ridge vs. Lasso regression.
 - AIC vs. BIC: Differences and applications.
 - Regression Equation: $y=a+bx$, where $a=0, b=0$. Explanation of what it means (not regression, as epsilon is missing).
 - Population regression line vs. Sample regression line.

AWADHESH MAHATO

Resume Shortlisting

Round 1:

Case: Selection of car, EV, petrol or diesel

Asked clarifying questions like what would be the usage for, how much is the timeline, etc.

Gave mechanical engineer's perspective on each vehicle types

Made cost breakdown in terms of fixed and variable costs of each vehicle type

Came up with price difference within which EV is preferred and beyond which Petrol would be preferred

Guesstimate: Number of taxis at a given time in your hometown

Proceeded with demand side approach

Made breakup of population and need for transportation vs modes of transports available

Case: Do sanity check of data of orders from Zomato to IIM C

Assumed the kind of features that would be made available

Would check for missing values, outliers and leverage points

In each feature would check for expected distribution

Puzzle: 9 balls, identical in appearance. 8 are of the same weight. 1 is heavier. Find the defective ball in minimum no. of measurements using a pair of balance scales

<https://www.mathsisfun.com/puzzles/weighing-9-balls-solution.html>

Behavioural Questions:

Elaborate your weaknesses.

Why this role?

Round 2:

Puzzle: Camel and Banana Puzzle

<https://www.geeksforgeeks.org/puzzle-15-camel-and-banana-puzzle/>

A person has 3000 bananas and a camel. The person wants to transport the maximum number of bananas to a destination which is 1000 KMs away, using only the camel as a mode of transportation. The camel cannot carry more than 1000 bananas at a time and eats a banana every km it travels. What is the maximum number of bananas that can be transferred to the destination using only camel (no other mode of transportation is allowed).

Approached by incrementally increasing drop point
Later found that it's already solved in GFG

Puzzle: Tapered circular Rod of some length need to find the midpoint, we can burn the rod from any end

Burn from both the ends simultaneously

Assumption: Rod consumption is directly proportional to volume

Himanshu Singh

Shortlisting

- Resume-based shortlisting.

First Round:

- Introduction (mentioned UPSC and Electrical Engineering).
- Asked about the current Power Minister and Power Secretary.
- Given a guesstimate to determine the electricity consumption demand of Noida.
- Asked clarifications on my approach.

Second Round: Puzzle

- Asked two puzzles. Camel Banana puzzle. Drug identification puzzle (test tube, rat, poison)
- Asked about my favourite Python library and why.

Third Round: HR

- Asked if I would join if given an offer.
- Some personal questions.
- Casual conversation.

JPMC

ARJUN KESRI

Position – CIB Research & Analytics Global research data science

(Using NLP to assist daily tasks)

Preliminary test (4 sections) – Negative marking is there – Moderate to Tough

Aptitude

Maths – Probability, P&C questions

Coding – C++(mostly OOPS concepts – Inheritance, encapsulation), Java MCQ's – Code was written and Output was asked

Both speed and accuracy are important

Coding - 1 hour given (challenging questions)

- 1) Array based – Pattern matching based on arrays, Pattern matching based on strings, we have to give output
- 2) Based on Binary Tree Data structure – we need to find distance from root node to leaf node of a tree. (Python 2.7 version is used)

(Many of syntax is different in Python 2 and Python 3. You need to be aware about syntax in both)

Suggestions – Basic DS and SQL on Hacker rank(array, stacks, Queues)

Round 1 (3 Panelists) (2 associates + 1 Executive director)

Intro

Randomly asking questions from each of projects based on keywords written

Process followed, results of projects, model inference? Did you do fine tuning? What did you do to improve results?

Asked to define the keywords of projects written

CFA/ FRM – Anyone with finance-based certifications will have an edge

Questions on portfolio optimization and Portfolio management – Asked about Sharpe ratio, Treynor ratio, S.D of portfolio of 2 assets?

Guesstimate – Estimate Number of Laptop users for work purpose in Delhi region?

Approach – Population of Delhi (Dividing them into office goers and Non-officers goers, Dividing further office goers into Tech based and Non-tech based)

Suggestion – You should form a team of 4-5 members and discuss a case/ guesstimate from casebook

Round 2

Retail price prediction (Linear Regression) – Nuances of techniques used?

Difference between Lasso and Ridge technique, working method behind it?

Questions related to Dataframes in Python on filtering, apply, map, zip functions (verbally we have to tell the code)

Suggestion – One should have thorough understanding about Pandas

Pandas – Python documentation

Data camp – Basic stuff

In SQL also, asked query to join tables and filter data. If you have 2 tables, How should be the structure of the tables? – verbally we have to tell the code

Poisson distribution – Probability of accidents occurring during halven hour given, Find probability of accidents in 1 hour period?

Puzzles related to Box and Balls – Some defective and good balls are given to you. What is the minimum number of balls you take to identify the number of defective balls?

Suggestion – Try to cover the Basic puzzles in Geeks for geeks

Round 3 (1 Panelist) – HR questions

General questions related to any background

Questions on culture and values of JPMC

Have you face any dilemma where someone close to you did a mistake and you stood on the right side?

Tried to check if candidate has understood what the role is about?

Suggestion – Be thorough with the Job description given

Round 4

Introduce yourself

Do you know about different verticals of JPMC?

What are different functions or areas in which JPMC works?

When and how you came to know about JPMC?

Suggestions – Read about companies on their website

SAHIL CHADHA

JPMC test: toughest. Quant questions MCQs - do standard quant books like this [book](#). Then there is one computer language MCQs round and maths MCQ round. 1-1.5 sections suffice.

Shambhawi

Round 1

1. Statistics Questions:

- Spearman Correlation Coefficient.

- Another correlation coefficient (unknown).
- 2. **Project Discussion:**
 - Questions related to projects on the CV.
- 3. **ML Basics:**
 - General questions on machine learning and statistics.
 - No deep learning questions.
- 4. **Guesstimate:**
 - "How many people in Mumbai travel by Uber in a day?"
 - Other approaches were discussed.
 - *Advice:* Refer consulting books for guesstimates and focus on the structured approach.
 - Approach: India → Mumbai → Urban/Rural → Voice out your thought process.

Round 2

1. **Master's Thesis Discussion:**
 - Topics included SVD and PCA.
2. **ML Concepts:**
 - Bagging, Boosting, and Random Forest.
 - ML basics and CV-related questions.
3. **Coding Question:**
 - What approach to take?
 - How to handle null values?
 - Use of dictionary and functions to solve problems.
4. **Puzzle:**
 - 3 jars (3 ml, 5 ml, 8 ml): Use them to measure exactly 4 ml in two jars.
5. **Behavioral Question:**
 - "Do you like Mumbai?"
 - Discussion about location preferences.

Aniket Sinha

Round 1

1. **Project Discussion:**
 - Favorite project: NLP project with LSTM-GRU (detailed discussion).
2. **Technical Concepts:**
 - LSTM structure, gates, and differences from GRU.
 - Regression: Algorithms in-depth, including mathematical concepts.
3. **SQL:**
 - Window functions: Lead, lag, rank, etc. (7–8 functions).
4. **Guesstimate:**
 - Problem: "How many people in Mumbai use Uber daily?"
 - Structured approach: India → Mumbai → Urban/Rural segmentation.

Round 2

1. **Master's Thesis:**
 - Topics: SVD, PCA.
2. **ML Concepts:**
 - Bagging, Boosting, Random Forest.
3. **Coding:**
 - Approach to handle null values using dictionary functions.
4. **Puzzle:**
 - Problem: Use 3 jars (3 ml, 5 ml, 8 ml) to measure 4 ml.
5. **Behavioral:**
 - Location preference: "Do you like Mumbai?"

Manas

Coding and SQL:

- Shared screen to write pandas data manipulation code.
- SQL queries.

Projects:

- Focused on linear regression as a "safe" project.

Behavioral:

- Discussed hobbies (philosophy).

HARSH RAJ

Round 1 (20 Minutes)

1. What does p, d, q mean in Time Series?
2. What is Box-Cox transformation?
3. Pandas data manipulation - Creating a DataFrame, merge, pivot table
4. SQL - Joins, Window function
5. Probability

Round 2 (20 Minutes)

1. Question from Portfolio Optimization project
2. Guesstimate - Estimate the number of toys sold by Hamleys in India daily

SUBHAV CHAUHAN

Pre Round:

The assessment included MCQs and coding questions covering Logic, DSA, and Algorithms. One of the problems involved basic string matching and substring retrieval. Another problem presented an array of heights for N pencil stands, with a cutter provided to trim them to the same height.

1st Round:

Panelist 1:

- Asked about L1/L2 regularization and its details.
- Inquired about multicollinearity and its implications.
- Discussed Decision Tree modeling concepts.
- Asked about Bagging vs. Boosting and their differences.
- Explored XGBoost and how it differs from other algorithms.
- Discussed Clustering and how to determine the optimal number of clusters.
- Asked about my project on the Recommendation System (Collaborative Filtering).
 - Specifically, he questioned how to backtrack a recommendation if only one recommendation is received per user.

Panelist 2:

- Asked SQL questions on JOINS and Rankings.
- Tested knowledge of Python (Databases and Pandas).
- Asked basic probability questions and a common Alice-Bob puzzle.

HR Round:

Why do you want to join this company?

Asked about my family background.

Asked about my previous work experience.

VARUN BHAGWANI

Test:

Few MCQs and 2 coding questions - easy DSA - alternate Knapsack type problem.

Round 1:

Statistical Questions:

Inference / Probability Related questions: T-test and Poisson distribution- how it will look?

Case Round:

How do you guess the number of iphones sold at a particular store just after it gets launched? Tell us using both the top down approach and the bottom up approach.

Project Related Questions - on my Stock Market prediction. What does decrease in RMSE represent?

Soumendu Mandal

Screening

1. CV-based shortlisting.
2. Coding test.

Interview Round:

- Customer Requirement Identification
- Stemming vs. Lemmatization—differences and applications.
- LDA
- Pandas-based join concat functions.
- 1 SQL question (basic level). Left join and select.
- Probability-related questions. Box has few blue balls , few red balls type prob ques

Sun Pharma

SUN PHARMA

HR Questions for 5 mins

Conflict based questions – If you are stuck in a conflict, How do you handle it?

What is your Daily routine?

Why are you interested in Pharma?

Asked about Projects – They asked questions on 3 Projects i.e., on Regression, NLP and CNN

Asked very simple that is about the model used and output of it?

Did you have any supply chain experience? Saurabh answered he has no supply chain work exp but he studied the supply chain elective during IIT Kgp semester. So, interviewer asked to explain the concepts studied in that subject.

Suggestion – If you are targeting Sun Pharma, Keep one supply chain project in the resume.

Saurabh has an opinion that Having supply chain experience will be helpful for this company.

RAJAN

Online – 25 min – 2 Panelists

Tell me about yourself?

Tell me about one of any projects?

(I mentioned regression)

Situation Based question – They explained dataset verbally which contain the sales data of different products and info about salesman like which sales member did what.

He asked to extract Top selling products and Top performing Salesmen from it. He wants whole approach of How I will do.

Some supply chain questions are asked – Don't remember them

HR Questions like what is your experience at PGDBA?

How's your experience as a fresher in PGDBA?

PRABAL

Round 1 (On Campus)

Where are you from? About my Background? I was preparing for UPSC, so they asked what if you decide to go back to UPSC?

Why Sun Pharma?

Asked about my Time series project? Where did you do this project? It is a very standard basic project, what else you could have done? (Ans – I said we can use exogenic variables)

They asked to explain about my fav project (Ans – Explained my regression project)

Guesstimate – They said you are from Chandigarh; Mohali is very close to Chandigarh and there is a cricket stadium there. They gave me a guesstimate on How many number of balls are required to cover the Mohali cricket ground.

Case – They are launching a drug for weight reduction. What are the factors you will consider in deciding its price? How would you market the product?

ARVIND

Non-tech roles – Manufacturing – work exp very important

Round 1 (Online)

TMAY

About your work exp

Have you done any projects of analytics in your work experience?

What previous companies do?

Choose 1 project – selected regression

Why did you Lasso? What is Lasso? Keywords?

Multi-collinearity? How do you tackle Multi-collinearity?

Focused on problem statement of the project? What Business problem do they solve?

Round 2

You have a unique profile

About work exp?

Which was the challenging project?

(Ans - Resume screening using NLP)

What this project does?

What is problem statement? What is solution?

Will your solution pass all the test cases?

HR questions

Why Sun Pharma?

Why Analytics after Food technology?

AKSHAT MISHRA

Round 1

TMAY

Questions based on the work experience – What was the basic tasks that were performed during my work exp?

Projects discussion – Sun Pharma doesn't ask candidates to pick their favorite projects. Instead they themselves pick several projects and cross question us based on the project statements written in CV.

Cross questions on Time series project, Finance project, Regression project.

What is your motivation to join sun pharma and more generic questions like that?

They value the candidates who know the vision, values, mission of sun pharma. In a settled way, they ask us about the Mission, values and Vision of Sun Pharma.

What is the project in which you faced the more difficulty? (I deliberately mentioned operations project, talked about few problems in that project)

Round 2 (Online)

Discussion in this interview was completely about Operations project. I explained all the nitigrities of this project.

Brief description about the Operations project - Inventory management – stocking analysis. How the stock is flowing in supply chain? When we have to replenish the stock? The dataset is based on the brewery stores. There were several brewery stores in the region and their inventory management was not good. Based on our data analysis, we have to make the plan to replenish the stocks on time. How the stocks are moving and rank the stores, vendors etc. Ranking the stores and vendors was the main highlight of our project)

Discussion about work experience – What I did and How I did?

KAPIL BHANWARIWAL

Round 1

Introduction

Where are you from?

Ans – Kota

Discussion about student suicides in Kota

They asked my hobbies

Ans – I mentioned Astronomy

How do you calculate the magnification of a telescope?

It is the ratio of focal length of objective lens and focal length of eye piece

If you are going to buy a telescope, what will you consider?

Ans – I answered I will consider aperture of telescope rather than magnification and gave the reasoning for it.

A question from medical insurance approval using Decision Trees from Scratch project – What is Type 1 error and Type – 2 error? I answered the definitions in terms of Null and Alternate hypothesis terms. But He had asked me to answer in the context of this project?

Soham

Round 1 (~10 min)

1. Why would you like to work in the Pharma industry?
2. Estimate the number of bike mechanics in Kolkata.
3. **Puzzle:** How many footballs would it take to cover an entire football field without showing any green?

Ramakrishnan

Round 1 (Work-Experience Heavy)

- **Panelists:** AVP + 2 HR representatives.
- **Focus:** Behavioral and situational questions.
 - Questions on client acquisition and challenges faced in previous work experience.
 - Situational scenarios requiring humble and innovative responses.

Round 2 (Cultural Fit Round)

- **Project Discussion:**
 - Choose and explain a project.
 - Emphasis on business problems and solutions involved.
 - Minimal cross-questioning during the discussion.

NITISH KUMAR

1. First Round:

- Focus on supply chain-related work experience.
- Case study: Identifying the optimal time to procure materials when prices are random.
- Guesstimate: "How many footballs can fit into a football field?" Missed accounting for the area where the coach stands.
- Overall: Went well.

2. Second Round:

- Conducted by a senior VP.

- Discussion on CV projects, how my previous work experience would improve post-PGDBA, and business acumen.
- Advice: Relevant work experience plays a crucial role.

Tata Trent (Waitlisted):

- Only Round:
 - Focus Areas: NLP text extraction, data pipelines, and model deployment.
 - Feedback: The interviewer seemed unclear about their expectations, leading to a less satisfactory experience.

Conclusion:

I am happy to have joined Sun Pharma, as it aligns well with my skills and experience.

SAMEER GUPTA

1. Shortlisting and Focus on Work Experience:

Shortlisting for Sanfarma interviews was based on CVs, with particular emphasis on having a background from IIT or NIT.

Work experience (especially in operations) was also a significant factor in selection, as some departments require candidates with operational understanding to manage business-related challenges effectively.

Since you're interning at Sanfarma, it's clear that operations experience is highly valued, especially in departments where business knowledge is crucial.

2. First Round Interview (30 minutes):

Introduction: The interview began with your introduction.

CV Review: The interviewers went through each line of your CV, asking for details about your academic journey, choice of PGDBA, and why you pursued Mechanical Engineering.

Project Deep Dive: They showed particular interest in your time series project, Gen AI project, and LLM-based project. Expect in-depth questions about these projects.

CFA: You were asked why you pursued the CFA certification and how it added value to your profile.

Linear Regression: The interview briefly covered the assumptions of linear regression, and then the focus shifted back to your projects. Most of the interview revolved around discussing your project work.

3. Second Round Interview (10-15 minutes):

The second round felt more casual and conversational.

Casual Questions: The interview with Srinivas, the Global Supply Chain Head, involved personal questions like:

How you spend your weekends.

Your marital status and future plans regarding marriage.

Whether you plan to pursue higher studies.

If you drink or not, and about your social life (e.g., girlfriend).

This round seemed more about assessing your cultural fit within the company.

Overall, the Sunpharma interview process is deeply rooted in your CV and work experience, particularly your projects and the operations background you bring. The first round is highly technical, while the second round leans towards personal and cultural fit.

SUBHAV CHAUHAN

CV based shortlisting.

1. Technical:

Discussion on my Regression project, mostly around Lasso/Ridge regression and when to use which.

Case Study - Oil Pricing- How will you decide on Oil Pricing - Time Series. At this moment should we purchase oil or not? Related factors.

2. Guesstimate:

Since you live in Haridwar, guess the total numbers of Tea Stalls in Haridwar.

3. HR:

What difficulties did you face in the last year? How did you overcome them?

Any Learnings.

Which institute among the 3 is your favourite and Why?

VARUN BHAGWANI

CV Shortlisting

Round 1: (Behavioural and Past Work Ex related questions)

Introduction

Your Strengths and Weaknesses.

Suppose you have been assigned a project to improve water facilities in a particular region. How will you do it? What kind of data will you collect? Long discussion on various aspects around this project.

Discussion on my clustering project on health insurance segmentation. I explained the dataset and what clusters represented?

Round 2: (IT Round - Techno Behavioural Round)

How do you go about learning languages as in coding languages?

When did you last face a problem and how did you overcome it?

Discussion on my credit default classification project.

Soumendu Mandal

- CV-based Shortlisting

Round 1:

- Introduction.
- Discussion on regression projects (not technical details, but why chosen and expected output).
- How to detect overfitting?
- How to explain overfitting to a layman?
- How do you ensure that model is not overfitting
- If you had to build a Data Science team, what characteristics would you look for in candidates?

Round 2:

- Asked about experience in Round 1.
- HR-based questions.
- Questions on RAG-based chatbot.

WWT

PRABAL

Round 1 (45-60 mins)

Covered regression, some basics of time series and CNN

In DL, we discussed about transformers, LLMs, RAG's, (we had a project on RAG's) CNN, PFT (parameter tuning), RLHF

Key query value calculations, Self attention (Basic concepts of transformers)

How can you explain self-attention to a 5-year-old?

One SQL question – Asked a query (On Group By)

Round 2

TMAY

Fav project? (Ans – Regression project)

Puzzles (2-3 standard puzzles from Greeks for geeks)

1 Puzzle – You have 2 containers of 3 ltrs and 4 ltrs. You have to measure 4 ltrs – (Refer GFG for exact question)

You have 9 balls – 1 is of different weight – You have to find the odd one out?

ARJUN KESRI

Round 1

TMAY

What projects have you done?

I answered Regression, Time series, NLP, NLP(Data structure), Deep Learning (2), GenAI (More focus on Gen AI project)

Basic working behind Generator and Discriminator? Architecture?

Puzzles/ Probability? (Q's related to cards)

Round 2

What WWT is currently doing? Do you know about that?

Questions on recent conference in which WWT participated

General discussion (Education background, what as your mindset behind choosing the course and reason behind applying for WWT)

Round 3 (HR Round)

Questions on projects in CV

Explain the problem statement, dataset?

What is the methodology followed during the project?

Business sense of project?

You are having the hot drink; you want that drink to cool down so that you can drink. How much time it would take to cool down?

Suggestion – Include projects related to Gen AI and DL (especially related to image based)

KAPIL BHANWARIWAL

Round 1 (25mins)

-TMAY

-Which is your favorite project?

Ans – Taxi fare prediction using Neural networks

-Interview asked questions in depth like what kind of activation functions you used? Why is it used? I developed 2 configurations, In first one, I used sigmoid activation. In second one, I used ReLU activation.

He has also asked me the calculations for forward propagation and backward propagation?

-I got better results in 2nd configuration where I used ReLU activation. So, He asked me what is the advantage of using ReLU over sigmoid?

Ans – I gave 2 advantages. I answered ReLU basically solves the problem of vanishing gradient. And also when you are calculating the derivative of activation function in backpropagation, ReLU is a simple function to calculate the derivative and calculation becomes faster and epochs can be performed faster.

I did one project one Countries clustering. This was from ML elective. So, I must write the algorithm from scratch. Interviewer asked me about What all algorithms I used. I answered K-means (Top down) and C-linkage (Bottom-up clustering algorithm). Interviewer seemed to have no idea about C-linkage clustering. So, he asked to explain the C-linkage clustering algorithm?

RAHUL SINGH

Screening – CV based Shortlisting

He directly picked the 2 projects RAG (Chatbot Assistant), NLP (Topic Modelling) from my resume, Asked me to explain the same & did a lot of cross questioning,

After that he asked to pick one more project to explain, chose regression, he asked one ques did ur model went underfitting & overfitting.

TIPS – This year it was very much clear that they gonna ask on RAG, NLP.

Surya Kiran

Round 1

1. Explain three deep learning projects.
2. Explain LDA (Latent Dirichlet Allocation) concepts.
3. What's the difference between Word2Vec and RNN?
4. Explain the transformer architecture and how it overcomes the drawbacks of Word2Vec and RNN.
5. Explain your regression project in layman terms to a client.
6. Solve a simple probability puzzle.

Round 2

- Repeated questions from Round 1 with additional case-based Python and SQL questions, such as merging three DataFrames with given attributes.

Round 3 (With Chief Data Scientist and India Head)

1. How would you handle multicollinearity in regression?
2. What ML/AI techniques could you have applied in your previous role in the automobile industry?
3. General questions about hobbies.

Verdict: Offered the role

Shambhawi

Project-Based Questions:

- Described and discussed 2-3 projects. Follow-up questions were asked on each.

Conceptual Questions:

- Bagging vs. Boosting:
 - Which one gets trained faster?
 - Pros and cons of each method.

SANGITA BHAKAT

Direct cv short list

Round 1 :

Intro

Asked to choose one project.

Basic discussion on regression project , subset of the questions asked in other interviews

Asked detailed questions on RAG project

What methods did we use?

What challenges did we face?

What could be done more on it ?

Retriever quality , how to check ?

Why OpenAI embeddings ?

What other embeddings can be used ?

Embeddings methods

Starting from one hot encoding to glove, Bert embeddings etc

Asked to explain the recommender system project.

Mathematical formulation of it .

I prepared it from Andrew ng video on recommender system.

HARSH RAJ

1. What is Text preprocessing, steps involved?
2. How does tokenization work?
3. From Regression project - What was the value of RMSE and MAPE achieved?

AWADHESH MAHATO

Resume Shortlisting

Round 1:

Explain NLP project.

Questions on terms related to those mentioned.

Explain differences in BERT, Bi-LSTM and GRU.

What is model quantization?

Explain Clustering project.

What is WCSS?

What's the logic behind the elbow method?

Explain each clustering algorithm.

PARTH PANDEY

Resume based shortlisting.

Round 1:

Introduction.

Discussion on my Gen AI project.

What do you mean by the accuracy of an architecture?

Discussion on my case competition where I had prepared a company's ESG table using RAG.

Madhur Rawat

Round 1:

- Puzzles.
- Basic questions on different ML models.
- Discussion on projects.
- Why WWT?

Round 2:

- Regression & Time Series concepts.
- SQL-related questions.
- Why WWT?

Soumendu Mandal

Screening

- CV-based shortlisting.

Interview Round 1:

- Questions on work experience.
- Puzzles. Point, go 10km south, go from here to there. Can you go to the same point?

Decision Point

ARVIND

Round 1

Where are you from?

TMAY

Time series project – Problem statement?

What is ADF test?

What is Durbin Watson test?

Complete flow of time project was discussed

Case was given – Point A to B, We have to classify whether a route is safe or not? Which all features are important to look at? Which model can be used?

Round 2

Time series questions of Round 1 are repeated

What is Adjusted Rsquare?

Questions on keywords in resume?

What is multi-collinearity?

Which all classification algorithms are there?

Stacking? Bagging? Boosting?

What activation function you use for classification?

What activation function you use for classification in case of multi-class classification?

Guesstimate – My interest was on football – You need to guess the total number of wins that happen in a year. (Ans – How many matches happen first need to be determined, then How many wins)

What is SVM? What is Kernel trick?

Round 3

HR questions

Why PGDBA after food tech?

Given a case – There is one company – 15-20 SKU's. Will you forecast the SKU's independently or combinedly?

Some product may be new? What will you do?

(ans – I told I will proxy data)

RAJAN

Round – 1 (Online) – 1 Panelist (25-30 mins)

Tell me about yourself

Scenario based question – He gave a scenario to me in which I must predict runs scored by the Rohit Sharma in next innings. What kind of features to be considered for this problem/scenario and the reasoning for that why I am considering this feature. What model do you use for this model and what assumptions do you consider?

What is overfitting and how do you tackle it

What is Bagging vs Boosting?

He asked me time series question. Every data point is giving same value as an output. What kind of phenomenon is this? Which model can be used to forecast this model of time series?

(Ans – It is stationarity, and simple moving average method can be used)

Round 2 – 30 mins

TMAY

Scenario based question – He explained me scenario in which I must find out which type of breast cancer does the patient has according to the size of tumor? – Multiclass classification problem – He asked about the approach, model and underlying assumptions that I will consider?

(Ans – I suggested many models starting from logistic regression in which multi class classification can be used and so on but he specifically anted How do use SVM for this problem

What is SVM and How it works and how it does multi class classification?

What are different models for classification

My elective was NLP. I was asked to give the brief history of NLP and How it evolved and asked what I learnt about NLP in IIT KGP?

Rahul Kumar

Round 1

1. Questions related to projects listed on the resume.
2. Time Series Project:
 - What dataset was used, why this problem, and what business problem is it solving?
 - What is stationary data?
 - Tests to check for stationarity.
 - Correlation vs. Autocorrelation differences.
 - ACF and PACF plots: Why are they used?
 - Why was the SARIMA model used?
 - VARMA: Why was it chosen?
3. Regression Models:
 - Assumptions of regression models.

- Concepts of heteroscedasticity and multicollinearity.
- 4. Bagging vs. Boosting: Explanation and differences.
- 5. Lasso and Ridge Regression: Applications and differences.
- 6. Questions on all keywords mentioned in the resume.

Round 2

1. Time Series Project: Follow-up questions (some repeated from Round 1).
2. Guesstimate: Count the number of railway stations in Bihar.
3. Data Science Case: Demand forecasting for multiple products.
4. CNN Project:
 - How is CNN used?
 - Transfer learning concepts.
5. Hyperparameter tuning in XGBoost.
6. When to use Ridge Regression vs. Lasso Regression.

Round 3

1. Business Problems:
 - Churn prediction problem: What business problem does it solve?
 - Write a pipeline for project deployment and retraining the model.
2. Advanced Topics:
 - What is data drift, and how do you handle it?
 - MLOps tools (Docker and Kubernetes): Do you know how to use them?
3. Focus: This round was more of a project fit round.

Cred

SAHIL CHADHA

Cred test: SQL question and coding question.

PRABAL

Online Test (1 DSA Question, 1 Data frame manipulation question, 1 SQL question on Hacker Rank)

Round 1 (Online)

They asked me about closed form equations, why we don't generally use Gradient descent in Linear Regression. What is the loss function? Panelists were expecting math behind all these concepts.

Panelists are expecting for coding use first principle (For loops etc.) instead of directly using Pandas and Numpy.

SUBHAJIT TARAFDAR

Round 1

Need to have very good understanding of Basic coding.

Code snake and Ladders game?

Whatever you do in statistics, you should be able to code.

Basics of Python, Numpy, Pandas, Matplotlib and DSA(Dynamic programming and Graph questions)

Asked about all projects on resume.

Regression-

Classification-

NLP – Draw the architecture of transformers?

ARVIND

(8-10 ppl shortlisted based on CV for Round 1)

Round 1

Keywords on resume (About CV) – 10-15 mins discussion

Decision tree or K-means

Asked to share the screen and code for K-means

Round 2 (1 hour)

All 7 projects were explained

All keywords were explained (and their counter words) – When I mentioned XG Boost, why not LG Boost?

Round 3 (Coding round) (Panelist – VP) (1 hour)

DSA – code (Better time complexity)

How NLP started? – History of NLP?

How NLP works

Manas

Round 1 (Very Intense)

1. **Resume Discussion:**
 - Detailed discussion of all bold terms.
 - Metrics like MAPE and classification/regression evaluation.
2. **Concepts:**
 - Non-linear regression: Why is it not commonly used?

Round 2 (Tanked)

1. **Technical and Coding:**
 - DSA knowledge was critical for this round.

Paras

Round 1 (1 Hour)

1. **Statistics:**
 - Central Limit Theorem (CLT).
 - Distributions (PDF, PMF, properties).
 - Simple probability questions.
2. **Time Series:**
 - Favorite topic: Basic questions were asked.
3. **Classical ML:**
 - Gini Index, Decision Trees.
 - Node conflicts in Decision Trees.
 - CART vs. C4.5.
 - Logistic Regression vs. SVM (differences).
 - Feature importance using SHAP values.
 - Objective function of XGBoost (Taylor approximation).
 - Tree construction in XGBoost and handling missing values.
4. **Deep Learning/NLP:**

- ReLU, Leaky ReLU, backpropagation.
- Word embedding techniques.
- LSTM vs. RNN, Skip Gram architecture.
- BERT vs. LSTM.

5. Case Study:

- Who should be referred to? How to determine if two phone numbers are "friends"?
- Features to take for a real project involving phone numbers.

Round 2 (1 Hour, Director Round)

1. Introduction:

- Personal introduction and thesis discussion (ISI Alumni).

2. Clustering Algorithm:

- Favorite algorithm: K-means.
- How to decide the value of k.
- Disadvantages of K-means and handling outliers.

3. Probability and ML:

- Medium-level probability questions.
- Standard ML questions.

4. Coding Simulation:

- Draw points from the unit circle and find the expected distance from the center.
- Monte Carlo simulation: Define and code the problem. The solution had to run successfully.

PARTH PANDEY

Test:

Data manipulation using Pandas and Numpy.

Questions from Statistics, ML and basic DSA questions on time complexity.

Round 1:

Very detailed and conceptual questions on Regression, Time Series and Statistics. He also asked about Hypothesis testing and t-test and when to use it.

He also asked about the Central Limit Theorem.

Round 2:

Questions on different projects, mostly data related.

Details about my Regression project.

He showed me a code and asked about its time complexity. I guessed $O(N^2)$. He asked me how to reduce it.

Tata Trent

ASWIN

(Checking If I can do Power BI)

Intro

What is DAX? Do you know Power BI? Some questions on Power BI?

RAJAN

On Campus – 2 Panelists (1 Technical person + 1 HR) – 25 mins

Tell me about yourself?

Tell me about one of any projects?

What is ridge regression and lasso regression? Which technique penalizes more?

As I mentioned in my project about Isolation Forest, So I was asked about What is Isolation forest and what is it used for?

My elective is NLP and one of my projects is NLP, He asked some questions about it and there were some cross questions on it.

I used BERT in the project. So I was asked What is BERT and How it works?

I was asked about Transformers (Ans – I have drawn whole architecture)

I was asked some GPT Models – chatgpt, gemini and llama. What type they are i.e., encoder, decoder or what

Rahul Kumar

Round 1

1. Case Question:
 - For a specific store, determine the optimal number of shirts of a particular type.
 - What kind of data is required?
 - What model would be used?
 - How and when to retrain the model based on data drift?
2. Limitations: Forecasting was not applicable due to the new nature of the project.
3. Favorite Project:
 - Asked to select a favorite project from the resume and explain it in detail.

SUBHAV CHAUHAN

Tell me about yourself.

Data related questions from my regression project.

Discussion on ChatBot, LLM, Attention Models, Transformers, Encoders and Decoders.

How will you use CNN to analyze an offline store? What factors will you consider, for example, heat maps from camera, number of footfalls etc.

Polycab

ASWIN

(Looking for personality)

T MAY

What have you done exactly in the work?

Pick any project? – I picked NLP Project

What is embedding? What is vectorization? What is token? What are the models you used?
Which library was that?

Rahul Kumar

Questions on projects apart from the most comfortable project.

Work Experience: General discussion.

ARIMA Models: Concepts and explanation.

CNN Model:

- Explain the architecture and write it down.

Advanced Topics:

- Hierarchical clustering.
- General AI project.
- LAURA optimizer.

Piramal Finance

SAURABH

Round 1 (Offline) – 2 Panelists (Anurag Shukla (Batch 8) + Bharat (Batch 5))

Question asked by Bharat

Questions on Python – Based on Pandas, about groupby in python?

A Data frame of a loan is given. 6-7 questions were asked on it. For Example, How many loans a person took?

Question asked by Anurag Shukla

ML Basics – 40 min – Asked about projects + python + Linear regression + Tree based model

What is multi collinearity? How to handle it?

What is VIF?

RMSE and R^2 – which is better for your project?

What is Precision and recall? Which one should we use in fraud detection?

If you have large number of columns in a data set, how would you reduce the size of columns?

What if the number of features is more than number of data points, what will you do in case?

Questions on the projects – Asked to explain about the projects done

Why did you get the need to use non-linear trees?

(I first used linear regression, Multiple linear regression and then XG Boost, So He asked this question)

XG Boost – How it handles missing values? Does XG Boost work on small dataset?

Difference between Working of XG Boost vs Random Forest?

Working principle of random forest – It does Row splitting or column wise splitting?

Round 2 – Offline - HR Round (1 hour)

There were some situation-based questions

If your team have a requirement of 1 person, 2 are talented males, 1 is female not so talented. Whom do you choose to be in your team? – Saurabh gave preference to talent over gender, so a follow up discussion about gender diversity had happened for 15 minutes.

Suggestion – Prepare situation-based questions well

ARVIND

Round 1 (Technical) (Panelist – PGDBA Batch 8 and Batch 5 Alumni) (20-25 mins)

Write syntax for joining 2 tables in Python?

Case was given – Multiple transactions happen in a month. I need to extract each day's outstanding balance of customers in a month? You were given 4-5 features. Balance outstanding, Date-time, Customer details (Customer name, ID), Transaction ID, Transaction mode. Code was asked in Python?

Adjusted R square? What is TFIDF?

Multi-collinearity?

What is Shiloth score?

What is word2vec? What is negative sampling in word2vec?

A case was given – How will you cross-sell a product. How will I find right customers to whom you can cross-sell it? – Used clustering

Round 2

2 Persons work for you in team. Team member wants to take leave in a crisis situation. Will you approve leave or not?

There are 2 Customers – Coconut vendor with income of 1 lakh per month or one with saloon shop with income of 20k per month

Conflict with senior manager – You came up with a idea, but your Senior manager want to follow his approach. You know your approach will give better results and manager is adamant. How do you convince him?

About my family background?

Paras

Round 1 (40–45 mins)

Panelists: Both were PGDBA seniors.

1. Pandas and ML Basics:
 - Standard pandas operations: grouping, merging, concatenation, and aggregation.
 - Case: Complaint queries moving between desks (desk 1 → desk 2, etc.).
2. Machine Learning:
 - Time Series:
 - Define time series.
 - Stationarity tests (ADF alternative).
 - ACF and PACF plots, ARIMA model equation.
 - Residual analysis.
 - Walkthrough of an end-to-end time series project (10–15 mins).
 - General ML:
 - BERT and XGBoost: Discussed usage.
 - Encoding and scaling: Explained types and methods.
 - How to encode columns with many categories (ordinal encoding based on proportions).

- Difference between LightGBM and XGBoost.

3. Statistics Questions:

- Hypothesis Testing, 2-sample test, KS test.
- Decision Trees and XGBoost: What makes XGBoost unique?

4. Case Study:

- Default prediction for the next 2 years.
- How to evaluate the model's performance.
- Handling data drift.

HR Round

- Behavioral:
 - POR (Positions of Responsibility) questions.
 - Challenges faced and how they were overcome.

L&T Finance

Dhananjay Kumar's Interview Experience :-

L&T converted

L&T

First we had, aptitude round, which consisted of english, maths, iq, coding. All were easy but coding was a bit tricky. You must have good knowledge of coding. One more thing in English it consists of antonyms, synonyms, active passive, odd one out, RC passage, inference questions like CAT. Overall it consists of the syllabus of CAT and SNAP but the level of difficulty is low.

Make Sure you need to clear sectional cut off of each subject only.

Interview

There were multiple panels. One panel is easy only asking basic ML questions but mine panel was a bit more oriented toward statistics. Luckily I was the only guy who was not asked any statistical question.

There were 2 people in my panel, one had proficiency in statistics and the other was HR. Both are senior people.

Interview question

- 1- Case Study : Suppose you are playing a rummy circle and you need to find a person who is winning the game not because of bias.
- 2- Assumption Linear regression
- 3- What is multicollinearity
- 4- What's the problem in multicollinearity
- 5- Suppose you have 3 columns C1, C2 and C3. C1 consists of 1,2,3,4. C2 consists of 1,4,9,16 and C3 consists of 1,8,27,64. Is there multicollinearity present in it?
This question is very basic if you are not able to answer it. Make Sure you work on basics.
I was not able to answer that. Later when the interview corrected me I got the answers correctly.
- 6- What is Rank, How will check multicollinearity by rank. (interview asked me to google it and find the answer and still I wasn't able to do it.
- 7- Till this time the situation was already tense.
- 8- Then he asked me about Live project Classification, then SVM, assumption of SVM, pros and cons on SVM.
- 9- He asked me about my favorite subject. I do not want to say math, statistics or any management subject. That's why I said clustering even though it is not a subject but I was very confident about it.

10- Case study – Suppose there are 2 people A and B belonging to country A and country B respectively. Find the minimum distance between A and B. Don't think of euclidean distance.

Think from the perspective of Clustering. There is a concept of soft clustering and hard clustering is going to be used.

HR question:

1 Why did you join PGDBA? Why PGDBA?

2 What's the difference between PGDBA and other b-schools?

3 what IIM Calcutta has taught you. (Be specific in your answer) . It was the only question, which I was given with full confidence. Your answer should be very simple to understand.

4 Why L&T , what do you expect from us ?

Finally converted

Be thorough with the answer. Make your answer very simple to deliver. Be prepared for HR questions. Most companies reject students in the HR round only .

Never Lose Hope !

UBER

Subhav Chauhan

Pre Round -

Probability related questions

Basic concepts like efficiency and effectiveness.

1st Round -

Introduction (Tell me about yourself)

How can we achieve optimization in dynamic pricing? Why ML model? How does it solve the problem? Elaborate.

Take in account all important factors like supply demand, weather conditions, traffic conditions etc.

This took around 20 mins of discussion.

TIP - Knowledge about Company's business model and operational strategy will be helpful.

AUXO AI

SUBHAV CHAUHAN

1st Round: Technical Round

Programming questions on basic optimization techniques.

Database question: How to normalize a table?

Discussion on XGBoost: Why is it so effective?

Discussion on Pretraining of LLMs.

2nd Round: Advanced Concepts & AI Discussion

Tell me about yourself.

Discussion on LLMs vs. SLMs: Are SLMs (Small Language Models) really small?

Fine-Tuning: Parameter-Efficient Fine-Tuning (PEFT).

Distilled Models: How does distillation work in LLMs?

Retrieval-Augmented Generation (RAG): Prompt engineering techniques.